

1 **The skeletal reconstruction of *Barosaurus lentus***
2 **in the American Museum of Natural History**

3

4 Keywords: *Barosaurus*; sauropod; neck; rearing; skeletal mount; cast

5 **Abstract**

6 The rearing mounted skeleton of *Barosaurus lentus* in the Roosevelt Memorial Hall of the
7 American Museum of Natural History (AMNH) is among the most culturally influential of all
8 dinosaur exhibits. We review the history of the genus and of AMNH 6341, the specimen on which
9 the mount is principally based. Excavated by Earl Douglass from the Carnegie Quarry at what is
10 now Dinosaur National Monument, this skeleton was divided between three institutions before
11 Barnum Brown reunited it at the AMNH around 1930. It was then largely neglected for six decades.
12 We document the conception of the rearing mount during the 1986–1991 renovation of the fossil
13 halls, when Lowell Dingus and Eugene Gaffney independently arrived at the idea of a rearing
14 sauropod inspired in part by Gregory S. Paul's painting *Ambush at Como Creek*, and the
15 construction of the cast by Research Casting International. At about 15 m, it is the tallest free-
16 standing mounted skeleton in the world. Crucially, the mount's purpose was not to assert that
17 *Barosaurus* habitually reared, but to provoke visitors to think about what fossils can and cannot tell
18 us about extinct animals. About 80% of the material in the mount was cast from AMNH 6341.
19 Almost all of the rest was cast from second-generation molds of the Carnegie *Diplodocus*, including
20 a third of the neck and some residual camarasaurid material in the manus. We summarise the
21 controversy the pose provoked, trace the longer history of rearing sauropods in science and art, and
22 assess the mount's substantial impact on popular culture.

24 Plain language summary

25 *Barosaurus* was a long-necked plant-eating dinosaur, a close relative of *Diplodocus*, that lived in the
26 American West about 150 million years ago. The skeleton behind the American Museum of
27 Natural History's famous mount was dug up by Earl Douglass in Utah in the 1920s and was at first
28 mistaken for a *Diplodocus*. Its bones ended up split between three separate museums, until Barnum
29 Brown bought all three portions for the AMNH around 1930. The skeleton then sat mostly ignored
30 for sixty years.

31 During a major renovation in the late 1980s, two museum staff independently hit on the idea of
32 showing a sauropod rearing up on its hind legs — an idea partly inspired by a Gregory Paul
33 painting. A Canadian company made lightweight casts and assembled them into what is still the
34 tallest free-standing mounted skeleton in the world, about 15 metres tall. Roughly 80% of the cast
35 comes from the real specimen; most of the rest, including part of the neck and the skull, comes from
36 *Diplodocus* casts.

37 The pose was never meant as a firm claim about behaviour, but as a prompt for visitors to ask what
38 fossils really can and cannot tell us. It sparked scientific argument — some of it in a medical journal,
39 about how such an animal could pump blood to its raised head — and later biomechanical
40 modelling suggested the pose was indeed feasible. The mount has since shaped how sauropods
41 appear in films, television and toys.

42

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44 **Introduction**

45 *Barosaurus* is a diplodocid sauropod from the Late Jurassic of North America, found in the extensive
46 Morrison Formation of the American West. It closely resembles its relative *Diplodocus* in most
47 respects but is characterised by an extremely long neck, even by sauropod standards. In the popular
48 imagination, it is typified by the iconic rearing mount in the Roosevelt Memorial Hall of the
49 American Museum of Natural History (Figure 1).

50 Although the material that the mount is based on (the partial skeleton AMNH 6341) has never been
51 described in monographic detail, the mounted skeleton has been enormously significant culturally,
52 and it is due to this that *Barosaurus* is universally recognised as proportionally long necked in popular
53 books (e.g. Bartram et al. 1983, Lindsay 1992, Lambert 2000). Along with the Carnegie *Diplodocus*
54 CM 84 and *Apatosaurus* CM 3018, and the Berlin *Giraffatitan* MB.R.2181, it has been one of the
55 keystone specimens in establishing the perception of sauropods by the general public.

56 There are two popular accounts of the *Barosaurus* mount (Norell et al. 1991, Dingus 1996:21–26) but
57 as yet no scientific account has been published. In this paper, we will review the history of *Barosaurus*
58 (Table 1). We will consider the composition of the mounted *Barosaurus* skeleton in the spirit of
59 Janensch's (1950) review of the original Berlin mounting of *Giraffatitan* (= "*Brachiosaurus*" of his
60 usage) *brancai*. We will determine which parts are cast from the main specimen AMNH 6341, and
61 which from other specimens. We will discuss how the pose was decided on, summarize the
62 controversy generated by the mount, survey the history of rearing sauropods, and consider the
63 cultural impact of the AMNH exhibit.

64 **Anatomical nomenclature**

65 *Cn* indicates the *n*th cervical vertebra, *Can* indicates the *n*th caudal vertebra.

66 We follow McIntosh's (2005) interpretation of the presacral vertebrae of *Barosaurus* consisting of 16
67 cervicals and nine dorsals, rather than 15 and 10 as in *Diplodocus* (although see discussion below).

68 **Institutional abbreviations**

- 69 • AMNH — American Museum of Natural History, New York, New York, USA.
- 70 • CM — Carnegie Museum of Natural History, Pittsburgh, Pennsylvania, USA.
- 71 • HMNS — Houston Museum of Nature and Science, Houston, Texas, USA.
- 72 • MB — Museum für Naturkunde Berlin, Berlin, Germany; specimen numbers for fossil
73 reptiles take the form MB.R.*nnnn*.
- 74 • ROM — Royal Ontario Museum, Toronto, Ontario, Canada.
- 75 • USNM — United States National Museum, Washington, D.C., USA.
- 76 • YPM — Yale Peabody Museum, New Haven, Connecticut, USA.

77 **Historical background**

78 **Early discoveries of *Barosaurus***

79 As recounted in McIntosh (2005:40–41), the first fossils of what is now *Barosaurus* were discovered in
80 the 1880s by Mrs. E. R. Ellerman on land owned by Mrs. Rachel Hatch half a mile east of
81 Piedmont on the eastern rim of the Black Hills of South Dakota. In the summer of 1889, O. C.
82 Marsh visited the site with J. B. Hatcher, and collected part of the tail, obtaining a promise from
83 Ellerman and Hatch that they would protect the rest of the specimen until it could be collected.
84 Based on six caudal vertebrae and a chevron from this initial excavation, Marsh (1890) very briefly
85 described and named the new genus and species *Barosaurus lentus* in a six-page paper in which he also
86 cursorily described the theropod *Ornithomimus* and two new species of *Triceratops*. The only *Barosaurus*

87 elements mentioned in Marsh's description were caudal vertebrae, and a single mid-caudal centrum
88 was illustrated (Marsh 1890: figures 1–2). Marsh's diagnosis noted only that the caudals resembled
89 those of *Diplodocus* but varied from them in several ways that subsequently turned out to be errors
90 brought about by comparing more anterior *Barosaurus* caudals with more posterior *Diplodocus*
91 caudals.

92 It was not until eight years later that Marsh attempted to have the rest of the skeleton collected,
93 sending George Wieland in late August 1898. In the intervening time, Mrs. Ellerman had died and
94 parts of the skeleton had been taken by locals, but Wieland was able to reunite much of this
95 material and excavate what remained underground, apparently working alone (Wieland 1920:529).
96 All the material was shipped to Yale and added to the holotype under the specimen number YPM
97 429.

98 However, Marsh died the next year, and work on the specimen stalled. Almost two further decades
99 passed before YPM 429 was fully prepared and Richard S. Lull was able to make a presentation on
100 the specimen at the end of 1916 at the eighth annual meeting of the Paleontological Society in
101 Albany, NY. Unfortunately, his abstract (Lull 1917), at only 74 words in length, is largely
102 uninformative. Subsequently he described the specimen in detail in a significant monograph (Lull
103 1919), which remained the definitive publication on *Barosaurus* until McIntosh's (2005) revision.

104 Since Lull's monograph, *Barosaurus* has become known from several additional specimens. These
105 include several excavated by Earl Douglass, working for the Carnegie Museum, at what is now
106 Dinosaur National Monument, north of Jensen, Utah. One of these specimens was broken up into
107 a cervical sequence CM 1198 (consisting of cervicals ?12, ?13 and ?16) and the postcervical skeleton
108 ROM 3670 — now reunited at the Royal Ontario Museum in Canada under the specimen number
109 ROM 3670. Also excavated by Douglass from Dinosaur National Monument was CM 11984,
110 another partial cervical sequence consisting of C7–C15 but still not fully prepared, residing the
111 collections of the Carnegie Museum.

112 Diplodocid material from the Tendaguru Formation of Tanzania was rather casually referred to the
113 new species *Barosaurus africanus* by Janensch (1922:464), but the complex nomenclatural history of
114 this species can be ignored for our present purposes, as it is now regarded as belonging to the
115 separate diplodocine genus *Tornieria* as the species *Tornieria africana*.

116 The most complete and informative *Barosaurus* specimen to date is AMNH 6341, the individual that
117 provided most of the material for the AMNH mount. It was briefly described as part of McIntosh's
118 (2005) revision of the genus *Barosaurus*, but has yet to be described in monographic detail (and will
119 not be so described herein). For the remainder of this paper, we will focus on this specimen.

120 **The AMNH specimen of *Barosaurus***

121 **The Carnegie Quarry at Dinosaur National Monument**

122 The fossil material of AMNH 6341 comes from the Carnegie Quarry at Dinosaur National
123 Monument, in the north-east corner of Utah. Earl Douglass, a fossil hunter employed by the
124 Carnegie Museum, received a letter on 6 August 1909 from the museum Director William J.
125 Holland saying that “he wished me to dig up Dinosaur bones east of Vernal”. (This and other
126 quotes in this section are from correspondence collection Douglass 1879–1953.) He and Holland
127 had seen a good *Diplodocus* femur on an earlier visit to the area but, arriving on 10 August, he found
128 that “someone had taken the best of the bones that were exposed”. On August 16 he wrote “The
129 bones were terribly broken up and it seemed as if they had been churned up ... Felt rather
130 discouraged.”

131 However, the very next day — the 17th and not the 19th as stated by Gilmore (1936:178) —
132 Douglass switched to another area and made an important find: “At last, in the top of the ledge
133 where the softer overlying beds form a divide, a kind of saddle, I saw eight of the tail bones of a
134 *Brontosaurus* in exact position. It was a beautiful sight ... several of the vertebrae had weathered out
135 and the beautifully petrified centra lay on the ground. It is by far the best looking Dinosaur prospect

136 I have ever found.” This skeleton, once excavated, would be mounted in the Carnegie Museum in
137 1913. It would be described by Holland (1915) as CM 3018, the holotype of the new species
138 *Apatosaurus* (= *Brontosaurus* of Douglass’s usage) *louisae*, and described in detail by Gilmore (1936).
139 As Douglass and his colleagues excavated, it quickly became apparent that there were many more
140 dinosaurs in the rock. Excavating these fossils now became Douglass’s full-time job (Figure 2). The
141 Carnegie Quarry, as it was named, would become one of the most fruitful in palaeontological
142 history. The Museum funded the excavations for 13 years, shipping 700,000 lb of material to
143 Pittsburgh (Gilmore 1936:170) — 318 tonnes.

144 **USNM work at Dinosaur National Monument**

145 By 1922 the Carnegie Museum had more fossils from this locality than it could use. Douglas
146 Stewart, Holland’s successor as Director of the museum, evidently favoured turning the quarry over
147 to the USNM, writing to Arthur S. Coggeshall on 6 September, 1922 to acknowledge the latter’s
148 recommendation that Douglass finish the excavation of a *Stegosaurus* and a small dinosaur
149 skeleton, but to leave two partially exposed specimens of *Diplodocus* to the USNM to excavate
150 (Stewart 1922a).

151 George Perkins Merrill, Head Curator of the Department of Geology at the USNM, learned from
152 Stewart that the Carnegie Museum was ready to relinquish its claim, and wrote on 6 November
153 1922 to William de Chastignier Ravenel, Secretary of the USNM, urging him to take on the quarry
154 as “There are few objects that would yield greater attractiveness to the main hall of this department
155 than the restored skeleton of one of the monsters of the dinosaur group that these beds yield”
156 (Merrill 1922). [By coincidence, the next day Stewart wrote to Merrill to confirm the offer and
157 provide details (Stewart 1922b).]

158 In a brief telegram on 10 November, Douglass reported to Merrill the existence of two sauropods in
159 the quarry (Douglass 1922a). The next day, Merrill wrote to Douglass requesting technical and

160 logistical information. Douglass replied on 22 November (Douglass 1922b) with a long and helpful
161 letter explaining how the two sauropod skeletons, both thought to be *Diplodocus*, were positioned —
162 #355 and #340, the former lying partly above the latter (Figure 3) — and stating that blocks
163 containing most of the neck of #340 had been taken out of the quarry.

164 Douglass listed 10 skeletons that had been partly or completely taken out during the year. Among
165 those listed was “No. 340 — *Diplodocus*. Apparently nearly complete skeleton but only neck taken
166 out. The remainder still in the ledge” (Douglass 1922c). By the end of the 1922 season, “all but two
167 skeletons — those of *Diplodocus* — were removed” (Douglass 1923). Work by the Carnegie crew
168 finished on 1 January 1923 (Osborn 1923).

169 Charles W. Gilmore arrived at the quarry on 14 May 1923 to work it for the USNM and was shown
170 around by Douglass, who was about to leave for Colorado (Gilmore 1923b). Gilmore inherited the
171 two partially uncovered skeletons, both then thought to be *Diplodocus*: #340, as mentioned in
172 Douglass’s report, and #355, presumably one of the two *Diplodocus* skeletons that Douglass had
173 referred to as not having been removed.

174 Douglass also handed over to Gilmore four large blocks and several smaller ones containing much of
175 the neck of #340, which as noted above had been extracted from the quarry but not prepared out.
176 It was Gilmore’s intention to extract #355 (the more complete of the two) for the USNM, then
177 secure the parts of #340 necessary to complete the skeleton (Gilmore 1923b). Skeleton #355 was
178 subsequently accessioned as USNM 10865 and mounted in Washington D.C. (Gilmore 1932),
179 where it remains in the public gallery to this day.

180 On 8 August 1923, Gilmore wrote briefly to Ravenel to say that work in the quarry was complete
181 and the extracted blocks would be shipped to the USNM soon after 19 August (Gilmore 1923a), and
182 sent a more comprehensive report in 30 August (Gilmore 1923b). In 1924, all excavation at the
183 quarry ended.

184 USNM's only interest in #340 was apparently in using elements of the skeleton to complete #355
185 for a gallery mount (Gilmore 1923b:1), so only the cervicals, three anterior dorsals, right scapula,
186 humerus and radius and a sternal plate were sent there (Gilmore 1923b:3). Some material of #340
187 went to the Carnegie Museum: Gilmore (1923b:1) wrote that "One block belonging to this
188 specimen was shipped to Pittsburgh, but will, I understand, be turned over to the National Museum
189 as soon as certain bones of another individual are extracted from it". It seems however that at least
190 *three* blocks went to Pittsburgh. One, containing part of the neck, was shipped from there to the
191 USNM in October 1924 (Coggeshall 1924); but there were also caudals, which would still be at the
192 Carnegie at the end of 1929: "nine or ten vertebrae in the Carnegie Museum are preserved in two
193 blocks 348/A and 348/B" (Brown 1929).

194 In April 1924, Earl Douglass was offered a position at the University of Utah and resigned his
195 position with Carnegie Museum (Douglass 1879–1953). He excavated the remainder of #340 and
196 sent it to the University of Utah. This left the skeleton now spread across three institutions in Salt
197 Lake City, Pittsburgh, and Washington, D.C. It is perhaps for this reason that, although the portion
198 of the skeleton at the University of Utah was prepared, it was never mounted.

199 The material of #340 in Washington was accessioned as USNM 11647 on 11 April 1927 (Abby
200 Telfer, pers. comm., 2025). When it was prepared, what had been thought in the field to be two
201 cervicals proved to be a single long cervical (McIntosh 2005:43). It was then apparent that the
202 cervicals (and the rest of #340) did not belong to *Diplodocus* after all but to *Barosaurus* (Gilmore
203 1932:4). This material could therefore not be used in the mount of USNM 10865, which was
204 instead completed with casts of the Carnegie *Diplodocus* CM 84 and unveiled in 1932 (Gilmore
205 1932).

206 **The skeleton reunited**

207 In 1929, Barnum Brown, acting for the American Museum of Natural History, visited most of the
208 nation's major natural history museums to assess their collections. He realised that the neck, anterior

209 torso and forelimb material at the USNM, and the tail segment still at the Carnegie, belonged to the
210 same individual as the partial skeleton at the University of Utah. Brown negotiated separately with
211 representatives of all three museums to acquire the three portions of this skeleton, and was able to
212 reunite the whole of Douglass's skeleton in New York at the AMNH, a museum that had had no
213 part in its excavation or early history. Brown arranged complex multipart deals: while the USNM
214 accepted a straight swap for their part of the *Barosaurus* with a skeleton of the tyrannosaurid
215 *Gorgosaurus* (AMNH 5428, now USNM 12814), the University of Utah made a cash-plus-fossils deal
216 in which they were paid \$2,500 cash plus the equivalent value in fossil mammal specimens (Brown
217 1929). The process of bringing the skeleton back together dragged on: the thirteen crates of
218 material at the USNM were transported to the AMNH only in February 1933 (Wetmore 1933),
219 perhaps because Wetmore himself — Secretary of the USNM — did “not favour disposal of this
220 *Barosaurus* neck” (Gilmore, reported in Brown 1929). The reunited specimen was given the specimen
221 number AMNH 6341. (See Norell et al. 1991:36–38, Dingus 1996:21–22, McIntosh 2005:42–43).
222 Brown's (1929) memorandum on the trading he did for fossils provided a list of *Barosaurus* material
223 at the USNM rather different than that which Gilmore (1923b:3) had specified when sending the
224 fossils there. Where Gilmore had listed 13 cervicals, three dorsals, right scapula, humerus and radius
225 and a sternal plate, Brown listed 10 cervicals, three anterior dorsals, left scapula and humerus. The
226 left/right differences for the appendicular material are simply errors on Brown's part (see below).
227 The radius and sternal plate mentioned by Gilmore seem to have gone missing: they are not
228 included in the material list below. They may still lurk unrecognised in the USNM collections. Matt
229 Caranno (pers. comm., 2026) thinks it is possible they are in the oversized collection at the remote
230 storage facility, but a detailed search would be laborious.

231 The issue of the number of cervicals is complex. Gilmore's count of 13 cervicals is compatible with
232 Douglass's (1922b:3) report that “the skull and two or three of the anterior cervicals of no. 340 had
233 been detached and were not in place” — an absence that would leave 13 cervicals remaining in an

234 animal with 15 of 16 neck vertebrae. The reduction to 10 in Brown's list, made after the material
235 had been prepared at USNM, can be explained as a consequence of supposed pairs turning out to
236 be single long vertebrae. This lower count is also in accord with the quarry map of Gilmore
237 (1932:figure 1; Figure 3). This shows nine dorsals and ten cervicals belonging to the *Barosaurus*
238 skeleton (although the line-drawings of the cervicals do not at all resemble the actual vertebrae:
239 compare Figure 4). If the cervical count of ten in the quarry map and Brown's memo is correct, the
240 most anterior of the cervicals (C7) seems to have been lost or destroyed some time in the next 30
241 years, since McIntosh's unpublished 1962 notes, his published account (McIntosh 2005) and the
242 present fossil holdings at the AMNH all include only nine cervical vertebrae, C8–16. Carl Mehling
243 (pers. comm., 2022) has searched in collections for the missing C7 and been unable to locate it. In
244 fact, the C7, if it ever existed, was likely lost or destroyed prior to the 1939 renovation: photographs
245 from that time (Figure 5B) show the anteriormost cervical vertebra on display, and it is recognizable
246 as the C8 that is the anteriormost currently preserved vertebra; and the signage that accompanied
247 the exhibit (Figure 6) indicates that nine cervicals were present.

248 **The AMNH mounted *Barosaurus***

249 **The conception of the mount**

250 The three parts of AMNH 6341 had been reunited from their layovers in the USNM, Utah
251 University and Carnegie Museum by 1933 or shortly thereafter. However, having acquired their
252 *Barosaurus* skeleton, the AMNH seemed at a loss to know what to do with it. It lay dormant for a
253 decade until the presacral vertebrae and some limb bones were exhibited in glass cabinets alongside
254 the mounted *Apatosaurus* in the Hall of Early Dinosaurs on 17 April 1939 (Figure 5, Figure 6). Plans
255 were made at that time to mount the skeleton in place of *Apatosaurus* (Anonymous 1939), but these
256 came to nothing. The *Barosaurus* elements remained here until the renovation supervised by Edwin

257 H. Colbert in the early 1950s. At this point, tentative plans were made once more to mount the
258 *Barosaurus* skeleton in the Hall of Early Dinosaurs, but it was felt that the mount would take up too
259 much space and these plans were abandoned. Instead, the entire specimen was moved into
260 collections. Four more decades were to pass before the skeleton (or at least a cast based on it) was
261 finally mounted.

262 In 1986, the museum began planning what would become an extensive renovation of its fossil halls,
263 which had become significantly outdated since the previous update more than thirty years
264 previously. The initial plan was to renovate only the Osborn Hall of Late Mammals, but a change of
265 museum leadership meant that by 1988 the project had become much more extensive, now
266 encompassing all four existing vertebrate fossil halls covering 65,000 square feet, and expanding into
267 new spaces at a total cost of \$38 million. Ralph Appelbaum Associates were hired as the designers.

268 As part of this broader initiative, paleontological staff were asked whether they had any specimens
269 suitable for mounting in the Roosevelt Memorial Hall that is the main entrance to the museum on
270 Central Park West. The hall, begun in 1931 and completed in 1936, is a majestic space in its own
271 right, with its impressive, barrel-vaulted ceiling. But it was puzzlingly empty as late as 1990 (see
272 illustration in Dingus 1996:20). It was the perfect space for a truly spectacular dinosaur mount that
273 could introduce new visitors to dinosaurs, draw them in to the main galleries, and provoke them to
274 think about paleobiological issues.

275 It occurred to Lowell Dingus, then project director of the fossil halls renovation project, that the
276 most spectacular exhibit would be a gigantic sauropod rearing up on its hind legs. Jack McIntosh
277 told Dingus about the long-neglected *Barosaurus* specimen in the collections, and it occurred to
278 Dingus that this could provide the basis for the mount. But he thought there was little chance of
279 persuading Eugene S. Gaffney, then the curator in charge of the dinosaur collections in the
280 Department of Vertebrate Paleontology, to undertake such a project. In fact, both Dingus and

281 Gaffney were known for their disdain of such speculative “dinomania”: the possibility of both
282 consenting to a rearing mount would have been considered very unlikely.

283 The iconoclastic palaeontologist Robert T. Bakker had in 1971 included a skeletal reconstruction of
284 a rearing *Apatosaurus* in an entry in the *McGraw-Hill Yearbook of Science and Technology* (Bakker
285 1971:figure 7f). This was provocative to the palaeoartist Gregory S. Paul, who incorporated the idea
286 in his 1978 painting *Ambush at Como Creek*. In this, only his third dinosaur painting (Gregory S. Paul,
287 pers. comm. 2022), he depicted a herd of *Diplodocus* surprised by an *Allosaurus*. As the carnivore
288 attacks, one *Diplodocus* provides cover for its retreating allies by facing down the predator in a rearing
289 threat display. This initial version of the painting was reproduced in Bird (1985:59), but Paul
290 became dissatisfied with it and painted over parts of the original in 1983 and 1985 to produce the
291 better known final version (Figure 7) in which the attack is by a whole pack of *Allosaurus*. This
292 version was reproduced in the influential book *Dinosaurs Past and Present* as Paul (1987:figure 16).

293 Knowing nothing of Dingus’s independently arrived-at plan, Gaffney found Paul’s painting
294 intriguing. Inspired by this artwork, he conceived for the Roosevelt Memorial Hall exhibit the very
295 ambitious idea of mounting a group of *Barosaurus* skeletons under attack from a group of *Allosaurus*.
296 Dingus was astonished to discover that Gaffney had come up with essentially the same plan as
297 himself — and both were further astonished when incoming dinosaur curator Mark Norell also
298 approved of the proposal, despite his own distaste for behavioural speculation about dinosaurs.

299 The original suggestion, using half a dozen or more skeletal casts, was deemed impractical, in part
300 because it would have taken up too much space even in the huge Roosevelt Memorial Hall. So while
301 the basic idea was adopted, it was scaled back to one erect *Barosaurus* adult and one juvenile, under
302 attack from a single *Allosaurus* — ironically, a scene corresponding more nearly to the original
303 version of Paul’s painting.

304 Dr. John (Jack) S. McIntosh, a professor of theoretical physics at Wesleyan University in
305 Connecticut, was an avocational palaeontologist specialising in sauropods, and through work done
306 in his spare time became the world's leading expert on the group. Gaffney knew him from his
307 undergraduate days and when he succeeded Colbert as curator of the AMNH fossil reptile
308 collection in 1970, McIntosh had provided a great deal of information about the AMNH dinosaurs.
309 He had told Gaffney that the museum had one of the best preserved sauropod skeletons in its
310 collection and that he thought it was the then poorly known *Barosaurus*. It would be the perfect
311 specimen to use as the basis of the rearing mount. Early in the design of the exhibit, Dingus and
312 Gaffney asked McIntosh what he thought about the pose. When he gave an enthusiastic "Yes, I do
313 think it was possible", the die was cast. Against their usual inclinations, Dingus and Gaffney had
314 become accomplices in the perpetuation of speculative dinosaur paleobiology, albeit for a
315 specifically limited purpose, as discussed in the next section. McIntosh was to have second thoughts
316 about the rearing pose, though. When interviewed a year later he observed "I'll just say this — I am
317 not responsible for the pose of the *Barosaurus*, and as a matter of fact, I would have been chicken
318 and would never have mounted it that way if it were my responsibility" (Psihoyos 1994:74), although
319 he did go on in the same interview to reaffirm that he thought the posture possible.

320 **The purpose of the mount**

321 While providing a breathtaking first impression of the museum for new visitors, the purpose of the
322 mount extended well beyond the visual power of viewing the tallest free-standing dinosaur mount
323 ever constructed. More importantly, the mount would introduce a major scientific theme that would
324 be integrated into all the fossil halls encompassed in the renovation project: what can we actually
325 know about extinct animals, and what can't we know based on the limited kinds of data preserved in
326 their fossils? The exhibition label for the mount clearly stated this conundrum in a section entitled
327 "Did this really happen?" (Figure 8):

No one knows for sure. Our only evidence of the lives of extinct dinosaurs comes from such fossils as bones and footprints. Fossils tell us about the size and shape of the animals and whether they stood on four legs (like *Barosaurus*) or on their two hind legs (like *Allosaurus*). They do not tell us, however, if *Barosaurus* could rear up or if *Allosaurus* hunted alone or in groups.

The creation of the mount

At the time of the casting-and-mounting project, the process was well documented with contracts, architectural drawings, inventories, shipping notes and other documents. But with the passage of more than three decades, nearly all this paperwork has been lost. The retirement of key individuals at the AMNH, including several authors of the present work, resulted in records being moved between offices and ultimately misplaced; and an extensive search of RCI's archives has turned up little. As a result, nearly all that follows has been assembled from the memories of the individuals involved, supplemented by information from contemporary and near-contemporary accounts such as those of Norell et al. (1991) and Dingus (1996), and the relatively few documents that were recovered from AMNH archives.

Early in the process, around March 1990, a prototype model at about 1/16 full size was commissioned at a cost of \$3,980 (Galiano 1990), to allow the scene to be created and manipulated at a workable scale (Figure 9). This was created by a model-maker at Maxilla and Mandible, a nearby fossil shop owned by Henry Galiano, which was one block north of the museum. It served as a general guide for the poses of the three animals, as well as a prop for raising funds for the renovation project. The model remains intact at the time of writing and was recently conserved. Comparing it with the finished mount, it is apparent that the torso is too long (it has one too many vertebrae and they are too elongate), the limbs are proportionally short, and there is a downward kink in the tail after the 7th caudal. The pose also differs from that of finished model in having a

352 much more strongly S-curved neck. Apart from these minor issues, however, it is an excellent
353 representation of the mounted skeleton.

354 Individual fossilized cervical vertebrae of *Barosaurus* can mass well over 100 kg, and supporting them
355 in the rearing pose would have required a prohibitively strong armature. Furthermore, permanently
356 mounting these scientifically significant fossils 10 m above ground level, even if logistically feasible,
357 would effectively make them unavailable for study. For these reasons, while the mounted skeletons in
358 the main Fossil Halls of the AMNH are mostly real bone, the Roosevelt Memorial Hall display
359 consists entirely of replicas. (The real presacral vertebrae are on display under the glass walkway
360 behind the mounted *Apatosaurus* in the Hall of Saurischian Dinosaurs, and have been since the Hall
361 opened in June 1995: see the photo in Dingus 1996:71.)

362 The bones of AMNH 6341, the *Barosaurus* specimen that was to provide most of the mount, were in
363 poor condition by 1990. The presacral vertebrae, scapulocoracoid and humerus had been in
364 collections for the best part of 40 years, since Colbert's early-1950s renovation; the rest of the bones
365 had been there for 60 years, ever since being reunited by Barnum Brown in 1930. Bones in
366 collections can degrade with time, especially the complex and delicate presacral vertebrae, and it is
367 not unknown for broken-off parts to become separated from the elements they belong to. A
368 program of repair and cleaning was required. As outlined below, about 80% of the skeleton was
369 present. These available *Barosaurus* elements were cast, and most of the remainder of the skeleton
370 was taken from a cast of the closely related *Diplodocus carnegii* with a few elements sculpted (details
371 below).

372 All casting and sculpting was done by Research Casting International (RCI), a recently established
373 organization specializing in mounting prehistoric animals. Founder Peter May had started working
374 with fossil mounts at the Royal Ontario Museum, beginning in 1977, where he learned the
375 techniques from paleontology technician Gordon Gyrmov and former WW2 Luftwaffe rocket-plane
376 test pilot Rudy Zimmermann. May rose to become the head technician at the ROM. Having

377 worked for a while at the Royal Tyrrell Paleontology Museum in Alberta, he found on returning to
378 the ROM that his expertise was in demand from other museums. Initially fitting this outside work
379 into his spare time under the banner of RCI, he went full time with his company in 1990, and the
380 AMNH *Barosaurus* was to become their most important early commission. May's first meeting with
381 Lowell Dingus, Gene Gaffney, Jack MacIntosh and Phil Currie was on 30 January 1990 at the
382 AMNH.

383 On 28 March 1990, an agreement was made between Lowell Dingus, acting for the AMNH, and
384 Peter May and Andrew Leitch, acting for RCI, to provide the fully mounted and painted cast
385 skeletons of a rearing adult *Barosaurus*, a juvenile, and an *Allosaurus* (Dingus 1990). The total agreed
386 cost was \$184,675 (equivalent to \$472,000 in 2026 according to usinflationcalculator.com). This
387 amount was made up of \$72,721 for the adult *Barosaurus*, \$75,136 for the juvenile, \$16,660 for the
388 *Allosaurus*, \$15,411 for a duplicate model of the juvenile *Barosaurus* for Dinosaur National
389 Monument, and \$4,748 for travel and transport. (Surprisingly, the small juvenile cost more than the
390 adult!). The agreement was signed by Dingus on 28 March, by May and Leitch on 30 March, and
391 by AMNH director William J. Moynihan on 3 April.

392 In the fall 1990, the RCI crew took the repaired and cleaned *Barosaurus* fossils in a semi truck from
393 New York to Toronto, where they remained for the best part of a year. There, the bones were
394 duplicated. First, they were coated with latex, which was then cured to form rubber molds (Figure
395 10). The bones were then cast, first with a layer of gel coat inside the molds, and then in various
396 materials. As noted by Norell et al. (1991:38), "Those that would stand near the ground were cast in
397 higher-density and more durable materials, while those higher up were made of lighter substances."
398 Specifically, the lower elements were cast in polyester resin and fiberglass with a substantial steel
399 armature, while the dorsal series, ribs and cervicals were cast with a much lighter fiberglass mat and
400 backfilled with polyurethane foam to keep the weight down as low as possible. (Due to the lightness
401 of these elevated elements, the steel armature could be reduced too.)

402 These accurate replicas of the original fossils, when painted, were indistinguishable from real bone
403 but weighed only a twentieth as much as the fragile and irreplaceable originals. The casts were not
404 “corrected” to account for distortion in the original bones: they are faithful reproductions of what is
405 preserved.

406 Most cervical ribs, however, were missing from the original cervical vertebrae. (These fragile
407 projections are rarely preserved intact.) The missing parts of the skeleton were mostly replaced by
408 elements cast from the Carnegie *Diplodocus*, though some were fabricated in Toronto. Records do not
409 show exactly which bones were sculpted but Peter May was the only sculptor at RCI at the time the
410 skeleton was being prepared, and he does not recall sculpting any elements other than the
411 metacarpals discussed below and possibly some cervical ribs.

412 In spring of 1991, a test erection of the rearing *Barosaurus* mount was carried out with the aid of a
413 hired crane and 15 m scissor lift (Figure 11). This had to be done in the parking lot behind the RCI
414 workshop, as the completed mount would be too tall to fit inside the workshop. The event was
415 attended by a group from the AMNH, and Jack McIntosh, who had been brought in as a consultant
416 to ensure that the bones were articulated correctly in the mount (Figure 12). Also present were
417 photographers including National Geographic’s Louie Psihoyos, a television crew, at least one
418 observer from another museum, and a crowd of local workers on their lunch breaks. The skeleton
419 was pieced together from prefabricated sections. The exercise began with the “tripod” of hindlimbs
420 and tail, anchored together at the pelvis, providing a stable base. These were followed by the torso
421 section, then the three sections of the neck and head, and finally the forelimbs. Dingus (1996:25–28)
422 recalled that “the strangest thing was that the mount actually looked rather natural and graceful
423 [...] I had always been extremely skeptical about whether sauropods could rear up to such heights,
424 but the grace of the mount almost erased my doubt”. Although the initial assembly of the rearing
425 barosaur had proven successful, another six months would be required before the entire exhibit was

426 ready to assemble in New York, as work was required not only on the main *Barosaurus*, but on the
427 juvenile and the *Allosaurus*.

428 To provide the base that the skeletons would be mounted on, fossil-bearing rock was considered
429 appropriate. During the summer of 1991, Gene Gaffney and Peter May searched for a suitable site,
430 finally finding an area that Gaffney was satisfied with by the road just outside the Fort Peck
431 Reservation in northeastern Montana. May and his crew later returned to the site and created peels
432 by spraying a thin layer of latex rubber across the rocks. They returned these to RCI, and used them
433 to make and paint a cast. Ironically, the exposures in this area are from the early Paleocene Fort
434 Union Formation (then the Tullock Formation), about 65 Mya, meaning that the ground that the
435 mounted *Barosaurus* stands on dates from after the extinction of the dinosaurs, about 90 million
436 years after the time *Barosaurus* lived.

437 In November 1991, the work in Toronto was complete, and Blakelock Moving were contracted to
438 ship the completed exhibit in sections for mounting to New York on 7 November (Clark 1991;
439 Figure 13). Unlike the trial mounting in Toronto, when the head and neck were attached in three
440 sections, in New York the three sections were first joined together and then fitted as a unit. This
441 large unit, about nine meters in length, proved awkward to manoeuvre, and it was difficult to raise it
442 to a sufficient height to slide into its slot (Figure 14). It took nearly two hours to wrestle it into place
443 at the front of the rearing torso. Worse, when the mount was finally completed it was imperfect
444 because, when the whole neck had been hanging horizontally, it had bent in the middle at the point
445 of suspension. The resulting kink in the neck can be seen between C11 and C13 in Figures 1 and
446 14B and in other contemporary photos such as those of Dingus (1996:26) and Lindsay (1992:18–
447 20). It was soon corrected, however, in a near-disastrous but ultimately successful late-night
448 operation (Dingus 1996:28), and the resulting line of the neck is smooth and elegant.

449 In a non-technical children's book simply titled, "*Barosaurus*", Lindsay (1992:14–19) provides a vivid
450 illustrated account of the molding, casting and mounting of the skeleton, with photographs by

451 Lynton Gardiner (see below). This book also contains the best published photographs of some of
452 the fossil material, e.g. a posterior cervical vertebra in left lateral view (p11), posterior cervical and
453 posterior dorsal vertebrae in anterior views and right pubis in lateral view (p12), articulated
454 sequence of cast dorsals and posterior cervicals in right lateral view (p18–19).

455 The exhibit — rearing adult and hiding juvenile *Barosaurus*, and running *Allosaurus* — was unveiled
456 to select guests on Tuesday 3 December 1991 (Anonymous 1991) and opened to the public on
457 Wednesday 4 December 1991 (Gordy 1991, Collins 1991). Press coverage was extensive, not only in
458 local publications such as *New York Newsday* (Gordy 1991) but also in national publications such as
459 the *Washington Post* (Associated Press 1991) and *Newsweek* (Begley and Yoffe 1991), and abroad in
460 countries including Germany and Japan.

461 It was at that time the only publicly exhibited *Barosaurus* in the world (Norell et al. 1991:36),
462 although additional mounts have since been erected at the Royal Ontario Museum in Toronto,
463 Canada, and the Natural History Museum of Utah in Salt Lake City. Also included in the exhibit,
464 in a case next to the mounted skeletons, were the real 13th cervical of AMNH 6341, and the partial
465 skull and neck AMNH 7530, which the juvenile mounted skeleton had been partially based on.
466 (That skull and neck, thought in the 1990s to belong to *Barosaurus*, were referred to the closely
467 related diplodocine *Kaatedocus siberi* by Tschoop et al. 2015:220). The juvenile skull and neck are, at
468 the time of writing, on exhibit in the Miriam and Ira D. Wallach Orientation Hall.

469 Since the mount of *Barosaurus* and *Allosaurus* opened to the public, tens of millions of visitors have
470 begun their visit to AMNH by sizing up this iconic, if somewhat controversial, skeletal scene.

471 **Photography of the mount**

472 During the mounting process at the AMNH, and after the unveiling, many photographs were taken.
473 But the rearing mount is a very difficult object to photograph well, climbing high into a gloomy hall
474 with bright windows. Among the photographers was Lynton Gardiner, who had been contracted by

475 Dorling Kindersley to provide photographs for the children's book *Barosaurus: On the Trail of the*
476 *Gigantic Plant-Eating Dinosaur* (Lindsay 1992) in their Dinosaur Spotter's Guides series, as part of a
477 broader contract to photograph dinosaurs at the AMNH and the USNM. Gardiner recalls (pers.
478 comm., 2022):

479 The main hall at the AMNH is huge, unevenly and dimly lit, so to get clear bright shots
480 I placed several 2400 and 5000 watt second Comet strobe packs and bare-bulb heads
481 around the hall, synchronizing them with slaves and an infrared sender on the camera.
482 With ISO 200 film we had a consistent f/11 aperture [enabling fine details to be
483 captured and light/shadow contrasts to be adequately depicted] from any vantage point
484 in the hall. I used a Hasselblad camera with a Distagon lens of about 40 mm. Except for
485 using digital cameras now, if I had the same assignment again I'd use a similar approach
486 with synced strobe packs around the hall. Available light with high ISO recording
487 doesn't produce the crisp detailed results you get with updated traditional electronic
488 flash lighting.

489 **The composition of the mount**

490 **Overview**

491 Norell et al. (1991:38) wrote that "only about a fifth of the skeleton was missing, but each of these
492 pieces, including the skull, several limb bones, and part of the tail, had to be modeled to complete
493 the skeleton [...] the technicians at Research Casting sculpted each individual missing bone in clay,
494 basing the shapes on the remains of more completely known close relatives of *Barosaurus*, in
495 particular, its contemporary *Diplodocus*." However, this account seems to be in error regarding
496 numerous elements being modeled from scratch. A complete list of the elements of AMNH 6341
497 itself can be assembled from the published works and notes of McIntosh, old signage, and letters
498 between Gilmore and the USNM (see below). Peter May recalls that with some exceptions discussed

below, the missing bones were filled in with casts taken from molds made from the Utah Field House's concrete copy of the Carnegie *Diplodocus*. That copy was made from the original Carnegie molds (Taylor et al. 2023), and the composition of the elements from which those molds were taken is well understood (Taylor et al. 2025), so it is possible to determine the sources of most of the bones in the *Barosaurus* mount. The details follow.

Barosaurus material in the mount

McIntosh's (2005:43) catalogue of elements in the referred specimen AMNH 6341 is incomplete and contains multiple errors. However, by reading it in context of the more detailed descriptions of the elements later in the same paper, and with his unpublished 1962 notes on the specimen, with observations of materials on exhibition in the past and present, and in light of historical correspondence, it is possible to arrive at a complete list of material as follows:

- the posterior part of the neck (cervicals 8–16)
- all nine dorsal vertebrae (dorsals 1–9)
- the complete sacrum (sacral vertebrae 1–5)
- the anterior part of the tail (caudals 1–29)
- six partial ribs, of which one is the first or possibly second on the left side, and another is probably its counterpart
- a single chevron, from further back than Ca13 and likely in the region Ca22–Ca28.
- right (not left as in McIntosh 2005:43) scapulocoracoid, fully fused together; distal end of left scapula; left coracoid
- right (not left as in McIntosh 2005:43) humerus

- right ilium complete except small part of the upper border; acetabular portion and “distal ends” of left ilium
- both pubes and ischia complete except parts of the proximal ends of both pubes and the right ischium
- complete right hindlimb (femur, tibia, fibula, astragalus)
- elements of the right pes: metatarsals I, II and V, phalanges I-I and V-I, two 2nd phalanges, one ungual
- three “ossicles” (McIntosh’s unpublished 1962 notes) — the meaning is unclear.

The sternal plates, clavicles and lower forelimb (ulna, radius, carpals and manus) are the only parts of the skeleton missing from both sides. Figure 15 shows a skeletal inventory of *Barosaurus lentus* with the portions of the skeleton preserved by AMNH 6341 in white and the missing bones in grey.

This list of material is gratifyingly congruent with that depicted in the 1939 exhibition panel (Figure 6). This panel provides the additional detail that the neural spines were missing from all five sacral vertebrae and caudals 1–3 and 26–28, and that only the anterior portion of caudal 21 remains. The panel shows 30 rather than 29 caudals: it is possible that some time after this illustration was prepared, the very fragmentary caudal 21 was lost or discarded, and that the caudals identified as 21–29 by McIntosh (2005) were in fact 22–30. The panel further suggests that two of the six preserved ribs may be those of the right side of dorsals 8 and 9. The only contradiction with the material list above is that the pes elements are interpreted differently: the panel shows metatarsals I, II and III, phalanges I-I, II-I and III-I, and ungual I. We consider McIntosh’s assignments more likely trustworthy.

This list of material of AMNH 6341 is mostly a superset of that listed as belonging to what was then the USNM’s part of the specimen in Brown’s (1929) account of reuniting the parts of the skeleton. The only additional element in Brown’s account is his statement that “the last ten cervical

vertebrae” (not just the last nine) were at that time present — see above. But Brown stated wrongly that the left rather than right scapulocoracoid and humerus are present (and the material list of McIntosh 2005:43 followed this error, contradicting his description and illustration later in the same chapter, McIntosh 2005:59–62). The girdle element that was on display in 1939 (Figure 5B) is clearly a complete and well preserved *right* scapulocoracoid, based on the location of the glenoid fossa in combination with the curvature of the shaft. Similarly the humerus that was on display in 1939 is the right, not the left, based on the shape of the proximal end and the anterior projections at the distal end. It is possible that Brown carelessly misidentified the elements in his letter and McIntosh transcribed the error in his 2005 paper. McIntosh’s unpublished 1962 notes further confirm the presence of these elements from the right (correct) side.

In preparing the mounted skeleton, casts of all the elements of AMNH 6341 were used. Little or no reconstruction was done, so distortion in the preserved cervical vertebrae is retained in the mounted copies; however, missing cervical ribs were added as necessary to the cervicals (Taylor, pers. obs.).

Non-*Barosaurus* material in the mount

So far as we have been able to determine, no casts of *Barosaurus* elements from specimens other than AMNH 6341 were used. The missing elements were filled in with *Diplodocus* casts that have a curious provenance. As recounted in detail by Taylor et al. (2023), the original Carnegie *Diplodocus* molds, having been used ten times by the Carnegie museum to create plaster casts from museums around the world, were donated in 1952 to the Utah Field House of Natural History in Vernal, Utah, and there used to create a concrete cast that was mounted outdoors (Untermann 1959). In 1989, Jim Madsen’s company Dinolab unmounted this concrete cast and used it to create second-generation molds with the immediate goal of providing the Field House with a new, lightweight cast to be exhibited indoors. These new molds were then used to create several additional *Diplodocus* casts. Jack McIntosh inventoried the AMNH 6341 *Barosaurus* material and ordered the missing bones from

568 Dinolab, as casts made from these molds, but we have not been able to find details of this order in
569 McIntosh's notes.

570 With this in mind, we will now consider the source of specific elements of the cast.

571 **Skull**

572 The skull in the mounted *Barosaurus* skeleton was cast from the corresponding elements in the
573 Carnegie *Diplodocus*. However, CM 84, the specimen from which the Carnegie mount is mostly
574 assembled, does not itself include a skull. As documented in Taylor et al. (2025:69), Holland
575 (1906:227) explained that the skull supplied to British Museum (now the Natural History Museum)
576 as part of the *Diplodocus* cast presented to it in May 1905 was a composite sculpture based on several
577 specimens. The posterior portion was modelled on material from CM 662 (now HMNS 175). The
578 remainder of the skull was based on USNM 2673, the skull on which Marsh (1896:175–179) had
579 primarily based his description of the skull of *Diplodocus*.

580 The skull used in the *Barosaurus* mount, cast from molds taken from this composite, is shown in a
581 1991 photograph (Figure 16A–B). It can be confidently confirmed as the same composite illustrated
582 by Holland (1906:figure 1) “as placed in the restoration at the British Museum”, and by Nieuwland
583 (2019:figure 5.3) in a photograph of a worker at the Muséum Nationale d’Histoire Naturelle, Paris,
584 France, with the plaster skull of their *Diplodocus* cast in 1908 (Figure 16C).

585 **Neck**

586 McIntosh (2005:45) considered the number of cervicals in *Barosaurus* to be 16 on the basis that there
587 are only nine dorsals, compared with ten in the closely related *Diplodocus*, and the most likely reason
588 is that the first dorsal was recruited into the neck. McIntosh’s inference has been widely considered
589 correct, and 16 is now the accepted cervical count for *Barosaurus*. (The earliest mention of the
590 sixteen-cervicals hypothesis in print appears in a caption in a children’s book, Lindsay 1992:20.)
591 Whether or not the AMNH 6341 material may at some point have included an additional cervical

592 vertebra (C7), at present it preserves the last nine cervical vertebrae. These are therefore considered
593 to be C8–C16.

594 The anterior part of the neck of the mount was completed using seven casts of anterior vertebrae
595 from the Carnegie *Diplodocus*, but the anteriormost seven *Diplodocus* cervicals were not used as that
596 would have resulted in an abrupt transition in length between the *Diplodocus* C7 (485 mm, Hatcher
597 1901:38) and the *Barosaurus* C8 (618 mm, McIntosh 2005:46). Instead, a non-contiguous sequence
598 of Carnegie *Diplodocus* cervicals was used to obtain a smoother transition. By comparison of
599 *Barosaurus*-mounting photos (Figure 17) with Hatcher (1901:plate III) and photographs of various
600 Carnegie *Diplodocus* mounts, the anterior cervicals cast from *Diplodocus* can be seen to be C3–6 and
601 C8 (512 mm) — and presumably the atlas and axis, though these are not in place in Figures 13 and
602 17.

603 To make matters more complicated, photographs show that the completed mount actually has 17
604 cervicals rather than 16 (e.g. Figure 1; Lindsay 1992:20–21; Taylor and Wedel 2016:figure 1).
605 McIntosh (2005:45) implicitly left open the possibility that *Barosaurus* had 17 cervicals, due to his
606 uncertainty about the correct interpretation of the 9th presacral, which he tentatively designated as
607 the first dorsal vertebra but considered possibly the last cervical. However, the existence of a 17th
608 cervical on this basis is predicated on a reduced dorsal count of eight, and the mounted skeleton has
609 nine dorsals with free ribs. The additional cervical therefore appears to be an insert not warranted
610 by the evidence available at the time of mounting.

611 However, photos of the mounting process in the AMNH (e.g. Figure 13) show that the neck section
612 used in the mount had only 14 cervical vertebrae — C3–16, not counting the atlas and axis which
613 were not in place at that time but which, when added, would bring the total to 16. The assembly
614 process was complex. The neck section containing the head and C1–16 was welded to the torso
615 leaving a stretch of bare armature. The two halves of the seventeenth vertebra (visible on the right
616 side of Figure 13) were then glued together around the armature, bridging the gap between the

617 bones of the neck section and the torso. The special 17th vertebra can be seen in Figure 11B,
618 distinctly darker than the cervicals above it and the dorsals below. Since there are no duplicate
619 vertebrae in the neck, the special C17 must have come from elsewhere. Careful comparisons
620 between photographs of the 17th cervical and 3D models of the Carnegie *Diplodocus* provided by
621 Heinrich Mallison (for the Berlin cast) and Paul Barrett (for the London cast) show that the former is
622 not a replica of any of the posterior cervicals of the latter. Since the Carnegie *Diplodocus* was the
623 only source of cast elements (other than the metacarpals discussed below), the 17th cervical of the
624 mounted skeleton must be a sculpture, crafted as an intermediate between C16 and D1.

625 (Note: it is quite possible that *Barosaurus* did indeed have 17 cervicals — or even more, as in the 18 of
626 *Mamenchisaurus youngi* (Ouyang and Ye 2002:24–30) or the 19 of *Mamenchisaurus hochuanensis* (Young
627 and Zhao 1972). Currently available *Barosaurus* fossils are not complete enough to determine the
628 point.)

629 CM 84, the specimen that forms most of the Carnegie mount, includes C2–8 (and the subsequent
630 cervicals), but not the atlas (C1). As discussed by Taylor et al. (2025:71–72), it is not clear which
631 specimen supplied the atlas in the Carnegie mount, and it seems most likely that this element was a
632 sculpture not based on any specific fossil specimen — and so the same would be true in the mounted
633 *Barosaurus*.

634 **Torso, sacrum and tail**

635 The complete dorsal sequence (of nine vertebrae) and complete sacrum (of five vertebrae) are
636 present in AMNH 6341. However, McIntosh's (2005:43) account says that only “six ribs and
637 fragments” were included from a total of 18 (two per dorsal vertebra), and as noted above the
638 identification of these ribs is uncertain.

639 AMNH 6341 includes the first 29 caudals (or perhaps caudals 1–20 and 22–30, see above) but only
640 one chevron. The missing caudals and chevrons were most likely provided once more by casts of the
641 Carnegie *Diplodocus*.

642 **Forelimbs and girdles**

643 The bones of the forelimb are not only absent from CM 84 (as noted by Hatcher 1901:45) but were
644 completely unknown in the Carnegie collection at the time the mount was constructed. Holland
645 (1906) made no comment on how the forelimbs were obtained for the skeletal mount sent to
646 England. As detailed by Taylor et al. (2025:84), both humeri, radii and ulnae of the casts produced
647 by the Carnegie Museum were sculpted based on the slightly smaller diplodocine individual CM
648 662 (now HMNS 175), which also provided the rear part of the skull (see above). The forefeet
649 however were sculpted from those of AMNH 965, now recognized as camarasaurid, and were given
650 three manual unguals instead of one, as well as too many phalanges. These errors were well
651 understood by the 1990s: to address them in the *Barosaurus* mount, a set of metacarpals from a left
652 manus assigned to *Diplodocus* (DINO 2568) was loaned from Dinosaur National Monument, cast and
653 returned. Corresponding sculptures were created for the metacarpals of the right manus (this
654 predated the use of scanning and 3D printing, which would have allowed a perfect reflection), and
655 the cast and sculptures were used in place of the camarasaurid metacarpals. Other elements of the
656 forefeet were taken from the Carnegie *Diplodocus* molds, but the unguals were removed from digits II
657 and III to correctly leave a single manual ungual. It follows that a small amount of camarasaurid-
658 derived material (the phalanges and surviving unguals) remains in the AMNH *Barosaurus* mount.

659 **Hindlimbs and girdles**

660 As noted by Taylor et al. (2025:84–85), the Carnegie Museum's own mounted *Diplodocus carnegii*,
661 uses the left fibula and partial pes of the referred specimen CM 33985, but for unknown reasons the
662 casts do not use this material. No documentation survives indicating what material was used to
663 create the left fibula and metatarsals III–V used in the casts. Most likely, these bones were mirror-

664 imaged sculptures of the right-side elements preserved in the *D. carnegii* paratype CM 94 (which also
665 furnished these bones on the right side of both the original mounted *Diplodocus* and the casts). And
666 the *Barosaurus* cast probably used second-generation casts of these sculptures.

667 **Discussion**

668 **Size of the AMNH 6341 animal**

669 The exact length of the neck of *Barosaurus* is difficult to determine as no complete neck is known.
670 Only one known specimen referred to *Barosaurus* preserves the anterior cervicals: AMNH 7535 is a
671 juvenile, consisting of cervicals 2–8, referred by Tschopp et al. (2015:220) to *Barosaurus* sp. Wedel
672 (2007:207) scaled these vertebrae up to match those of AMNH 6341 (C8 is preserved in both
673 specimens), to arrive at his total neck length estimate of 8.5 m. It seems that someone performed a
674 similar scaling operation using these vertebrae during the period of the mounting, as shown by notes
675 hand-written around 1990 on a printed draft of what would become the table of measurements in
676 McIntosh's (2005) *Barosaurus* paper (Peter May, pers. comm. 2022). The identity of the note-taker is
677 not known, but the handwriting does not match that of McIntosh himself. Summing the known
678 centrum lengths of AMNH 6341 cervicals 8–16 from this table (McIntosh 2005:table 2.1) yield a
679 total of 6993 mm. The scaled-up centrum lengths of AMNH 7535 cervicals 2–7 written onto the
680 manuscript are 125, 174, 234, 299, 355 and 467, for a total of 1654 mm. Together these sums add
681 to 8587 mm, a good match for Wedel's (2007) estimate of 8.5 m, which is currently the generally
682 accepted figure. However, see Taylor and Wedel (in prep.) for alternative estimates.

683 The height of the mounted *Barosaurus* is usually given rather inexactly as “fifty feet above the
684 Rotunda floor” (Norell et al. 1991:39), “almost fifty feet” (Dingus 1996:25), “five-storey-high”
685 (Gordy 1991:3) or “over 50 feet (15 m) from ground to head-level” (Lindsay 1992:26). Although
686 vague, these measurements are enough to establish it as the tallest mounted skeleton of any animal

687 anywhere in the world, about two meters taller than the Berlin brachiosaur which has “a skull
688 located more than 13 m above the level of the feet” (Remes et al. 2011:309). (The Berlin
689 brachiosaur was also mounted by RCI — or rather, remounted in 2005–2007, following the original
690 mounting in 1938 under the direction of Werner Janensch.)

691 The mass of *Barosaurus* AMNH 6341 is generally considered to be similar to that of *Diplodocus*
692 *carnegii*, as its torso is about the same size and the longer neck is balanced by *Diplodocus*'s longer tail.
693 Seebacher (2001) gave a mass estimate of 20,040 kg based on graphic double integration (compared
694 with 19,655 kg for *Diplodocus carnegii*). Henderson (2013) estimated 15,800 kg by slicing a digital
695 model (compared with 11,900 kg for *Diplodocus carnegii*). Benson et al. (2014), in supplementary
696 information S1, gave an estimate of 13,164 kg for AMNH 6341 based on limb-bone measurements
697 (compared with 13,801 kg for *Diplodocus carnegii*).

698 **Rearing pose**

699 The mounted *Barosaurus* is in a spectacular rearing pose, as though to defend its offspring against a
700 threatening *Allosaurus* individual. The exhibit was illustrated by a specially commissioned John
701 Gurche painting (Figure 18), which was used in gallery signage and in numerous publications (e.g.
702 the cover of *New York Newsday* for 29 November 1991; the cover of the AMNH's own magazine
703 *Natural History* for December 1991).

704 The unveiling of the Berlin cast of the Carnegie *Diplodocus* in 1908 had provoked controversy: Hay
705 (1908, 1910) and Tornier (1909) argued that its erect-legged posture was incorrect, and it should
706 sprawl like a lizard; and Holland (1910) emphatically rebutted these suggestions. In the same way,
707 the AMNH *Barosaurus* mount started discussions. From the moment of its unveiling this exhibit was
708 controversial for two reasons.

709 First, there is no direct evidence that *Barosaurus*, or any sauropod, practiced parental care (Sander et
710 al. 2008). However, it is well established from both trackways (e.g. Day et al. 2004) and death

711 assemblages (e.g. Coria 1994) that sauropods did live and move in herds of different-sized
712 individuals, whether genetically related or not. It is not unreasonable to assume that larger
713 individuals defended the smaller from attack on occasion.

714 Second, and more seriously, some paleontologists felt that *Barosaurus* could not or would not have
715 adopted the rearing pose — something which, as noted above, Dingus, Gaffney and McIntosh all
716 had their own reservations about. In a newspaper report published five days before the exhibit was
717 publicly unveiled, it was claimed that “of six leading paleontologists interviewed for this article, all
718 but one questioned how a behemoth weighing in excess of 25 tons could be accurately depicted in
719 an upright position. Most thought it physically impossible” (Gordy 1991:3) — although since the
720 article also wrongly claims that the posture was chosen “over the objections of Gene Gaffney”, it
721 should not be assumed to be accurate in other matters. While Kevin Padian was quoted supporting
722 the posture, Paul Sereno, Jack Horner and Phil Currie all expressed reservations — though none of
723 them went on to express their criticisms in scientific publications. Thirty years on, Sereno comments
724 “I still think it's ridiculous” (Paul Sereno, pers. comm., 2022), but Horner has mellowed: “I had
725 opposed the idea originally but have since come to the conclusion that at least the males had to have
726 been able to rear up to at least the back of the female. I think it took me about a decade to figure
727 that out. [...] I like the mount now, it may have been the way the males displayed. Would have been
728 spectacular!” (Jack Horner, pers. comm., 2022).

729 More specific criticisms and comments were to follow in published articles. Surprisingly, much of
730 the published discussion was in the medical journal *The Lancet*. First, provoked by Lillywhite's (1991)
731 brief account of the high blood pressure required to circulate blood to the head of *Barosaurus*, Choy
732 and Altmann (1992) argued that the great size required of the *Barosaurus* heart to perfuse an erect
733 neck would have meant that it beat very slowly. Check valves would therefore have been necessary in
734 the neck arteries to prevent the column of blood from falling back to the heart during diastole. They
735 suggested that these valves may have been active pumps — three pairs of “secondary hearts” —

736 though they admitted that in the absence of soft-tissue fossils this is pure conjecture. Millard et al.
737 (1992) dismissed this proposal as unsupported by evidence, noted the absence of check valves in the
738 neck of the giraffe, and argued that the multiple-heart scheme would probably create non-
739 continuous blood flow at the brain. They considered amphibious habits a more likely adaptation for
740 supporting the neck, despite the discrediting of that notion (e.g. Coombs 1975). In a letter published
741 simultaneously, Rewell (1992) noted based on dissection that while there are no valves in the giraffe
742 jugular vein, “specialised series of valves prevented reflux into tributaries running from below into
743 large veins”, a system also seen in the Bactrian camel and by implication also possible in sauropods.
744 Dennis (1992) claimed to have independently reached the same multiple-hearts conclusion as Choy
745 and Altmann (1992), and noted that hagfish, the giant Brazilian earthworm and some insects have
746 multiple perfusion organs. He also considered and discarded the idea that the blood flow to the head
747 operated as a siphon. In a letter published simultaneously, Taylor (1992) pointed out the evidence for
748 sauropod terrestriality, *contra* Millard et al.’s proposed amphibious habits, and that breathing would
749 have been impossible when the lungs were far below the surface (see Kermack 1951). And in a third
750 letter, Hicks and Badeer (1992) argued that the circulatory system was closed, i.e. not exposed to
751 ambient pressure, reducing the need for high blood pressure. (Badeer and Hicks (1996) would revisit
752 their ideas in more detail four years later, proposing a siphon loop circulating blood to the head and
753 arguing that blood pressure in the head was negative.)
754 Separately from the *Lancet* discussion, Landry (1992) argued briefly that *Barosaurus* would not have
755 had the necessary muscle mass to rear up, that compressive stresses in a vertical spine would exceed
756 the bending stresses in a horizontal spine, and that dropping down from a rearing posture would
757 create too much kinetic energy to disperse.
758 All discussion of this nature was welcome to the AMNH team. As noted above, the most important
759 purpose of the mount as originally conceived was not to argue in favour of rearing as a habitual
760 behaviour, but to set out the issue of what we can and cannot know. Dingus (1996:28) noted that he

761 actively welcomed the controversy — not only because all publicity is good publicity, but because
762 the mount's critics were encouraging the public to think about the controversy for themselves.

763 **A short history of rearing sauropods**

764 The notion of rearing sauropods has a long heritage. One of earliest life restorations featuring
765 sauropods is Charles R. Knight's 1897 drawing, created under the supervision of E. D. Cope,
766 appearing in Ballou (1897:20) and reproduced in Osborn and Mook (1921:figure 127). This shows
767 several *Amphicoelias* individuals in mostly submerged rearing postures.

768 Scientific support for the plausibility of rearing on land goes back at least to Osborn (1899:213),
769 who wrote that the tail of *Diplodocus* “functioned as a lever to balance the weight of the dorsals,
770 anterior limbs, neck, and head, and to raise the entire forward portion of the body upwards. [...]”
771 Thus the quadrupedal Dinosaurs occasionally assumed the position characteristic of the bipedal
772 Dinosaurs — namely, a tripodal position, the body supported upon the hind feet and the tail”.

773 In his classic monograph of *Diplodocus carnegii*, Hatcher (1901:58) strongly implied, without quite
774 explicitly stating, that *Diplodocus* habitually reared: “the modified nature of the chevrons of the mid-
775 caudal region indicate the point of contact of the tail with the earth attending the different positions
776 habitually assumed during the life of the individual”.

777 Riggs (1904:245–246), contrasting his new dinosaur *Brachiosaurus* with the then-known diplodocids
778 wrote: “We may well assume, with other writers, that the heavier forms, such as *Apatosaurus* and
779 *Diplodocus*, which are provided with long spines in the sacral and posterior dorsal region, were
780 adapted to rearing up on the hind legs as is represented in the conventional mounted skeleton of
781 *Megatherium*. In these forms we find that the body is short and therefore well adapted to this habit.”

782 Knight was painting rearing diplodocids as early as 1907 (reproduced in Taylor 2010:figure 6B).

783 Huene (1929:497) provided what was probably the first life restoration of a titanosaur, "*Titanosaurus*"
784 (= *Neuquensaurus*) *australis*. As well as the foreground individual, depicted in a classic four-square pose,
785 two more individuals are shown fighting in the background, one of them rearing on its hind legs.

786 Borsuk-Bialynicka (1977:51) proposed that the robust titanosaur *Opisthocoelicaudia* occasionally stood
787 on its hind limbs: "the tail working as a prop seems [...] probable. [...] Occasional bipedality is
788 therefore strongly suggested. It is, however, not a matter of briefly assuming a bipedal stance as is
789 possible in most of the tetrapods. [...] The animal must have been stable in this position; otherwise
790 this posture would not have left such distinct traces in the skeleton architecture."

791 In 1984, Gregory S. Paul drew an assemblage of four Jurassic sauropods: *Camarasaurus*, two
792 *Barosaurus* and *Apatosaurus*, all but the first of which were shown rearing to near-vertical postures.
793 This drawing, possibly the first artwork to show rearing *Barosaurus*, was reproduced by Bird
794 (1985:14).

795 In a landmark paper on dinosaur biomechanics, R. McNeill Alexander (1985:9) took the first steps
796 towards numerically modelling the ability of sauropods, including *Diplodocus*, to rear. These early
797 calculations were concerned only with the location of the centre of mass, and the ability of the
798 dinosaurs in question to shift it backwards to the location of the hips.

799 Robert T. Bakker had been arguing for some time that at least some sauropods reared (e.g. Bakker
800 1971, 1978). In his widely read popular-science book *The Dinosaur Heresies*, Bakker (1986:190–192)
801 reiterated and expanded his own argument that diplodocids, including *Barosaurus* itself, not only
802 could rear but habitually did so when feeding and at other times. The frontispiece of the book
803 illustrated this idea with a striking drawing of two apatosaurines fighting in tripodal posture. While
804 the idea of rearing sauropods had been growing among researchers, *The Dinosaur Heresies* probably
805 marked the point where the idea began to impinge on non-specialists.

806 Paul (1987:31), in the article containing his influential painting *Ambush at Como Creek* (see above),
807 argued that habitual rearing into tripodal stance is the only explanation for diplodocids' massive
808 hips and vertebrae, larger than those of otherwise bigger sauropods, and for their relatively straight
809 necks, short forelimbs and sledlike chevrons. The next year, his book *Predatory Dinosaurs of the World*
810 contained a drawing a *Mamenchisaurus hochuanensis* rearing to fight theropods (Paul 1988:88) in a
811 composition reminiscent of *Ambush* but rather more extreme due to the extraordinarily long neck of
812 *Mamenchisaurus*.

813 Jensen (1988) described a new camarasaurid species *Cathetosaurus lewisi* (considered by McIntosh et
814 al. (1996) to be a species of *Camarasaurus* but regarded as its own genus by Tschopp et al. 2014).
815 Jensen (1988:124–128) considered several characters of the *C. lewisi* holotype BYU 9047 as
816 specializations for rearing and bipedal locomotion: bifurcated neural spines through the entire
817 presacral series, elongated chevrons, a strong system of ossified diagonal intervertebral ligaments in
818 the dorsal vertebrae, supracostal plates in the sacral vertebrae, and 20-degree forward rotation of
819 the ilia relative to the sacrum. While some of these features do not seem directly related to rearing
820 and others are likely age-related, the rotation of the ilium remains unique and is difficult to interpret
821 as anything other than a rearing adaptation.

822 In more recent times, biomechanical modelling has been used to establish the feasibility of elevated
823 postures such as that of the AMNH *Barosaurus*. Mallison (2011) argued compellingly from kinetic–
824 dynamic modelling that diplodocines such as *Barosaurus* were particularly well adapted to bipedal
825 rearing and sustained tripodal (tail-supported) standing. So the pose selected for the AMNH mount
826 seems biomechanically justified.

827 **Cultural impact**

828 As well as catalyzing a scientific debate on the mechanical difficulties of rearing, the AMNH
829 *Barosaurus* mount has significantly influenced the depiction of sauropods in popular culture.

830 Pre-production of Steven Spielberg's *Jurassic Park* (1993) began in July 1990, and filming began in
831 August 1992, so the AMNH *Barosaurus* was unveiled about two thirds of the way through pre-
832 production, in time to influence decisions about the film's content. Notably, the first dinosaur seen in
833 the film is a rearing sauropod — but it is a *Brachiosaurus* rather than a *Barosaurus*. The choice of a
834 rearing *Brachiosaurus* in *Jurassic Park* is surprising because brachiosaurs were the most front-heavy of
835 all sauropods, and so the least well adapted to rearing. It must be possible that from early versions,
836 the script called for a *Brachiosaurus*, but the rearing was added only relatively late in the process, in
837 response to the AMNH exhibit.

838 If *Jurassic Park* was the most important representation of dinosaurs in film, the BBC documentary
839 *Walking With Dinosaurs* (1999) was certainly the most influential depiction on the small screen. The
840 second of the six episodes, *Time of the Titans* (broadcast in the USA as *Jurassic Giants*) traces the life of
841 a female *Diplodocus*, and spends much of its time with the herd of different-aged *Diplodocus*
842 individuals. Two passages of this episode show several individuals rearing onto their hind limbs: 15
843 minutes in, to push over trees; and 26 minutes in, to confront an *Allosaurus*, in a scene that can only
844 be a tribute to the AMNH exhibit. In this latter sequence, the rearing *Diplodocus* is even seen taking
845 several bipedal steps. More than 20 years later, Apple TV+ provided a spiritual successor to *Walking*
846 *with Dinosaurs* in *Prehistoric Planet* (2022). Episode 2 (*Deserts*) included a lengthy scene beginning at four
847 minutes, in which *Dreadnoughtus* titanosaurs reared into bipedal poses, initially for sexual display, then
848 for intraspecific combat.

849 The Dinosaur World park in Plant City, Florida, contains a life-sized sculpture of a rearing
850 *Barosaurus*.

851 Some dinosaur toys have been influenced by the AMNH mount: for example, the Collecta *Diplodocus*
852 is in a rearing pose. However, toys of *Barosaurus* itself are for some reason extraordinarily rare. The
853 Dinosaur Toy Blog (Smith 2007) has reviewed well over 2,500 dinosaur toys (Adam Smith, pers.
854 comm., 2024), including 50 versions of *Brachiosaurus*, 38 of *Apatosaurus* and 22 of *Diplodocus* — but

855 not a single *Barosaurus* toy. The only *Barosaurus* “toys” on the market at the time of writing seem to
856 be large museum-quality maquettes costing \$500 or more. A Lego model of the rearing AMNH
857 *Barosaurus* has been created by Ambrosino (a pseudonym) and its instructions made freely available
858 (Ambrosino 2023; Figure 19).

859 **Conclusion**

860 It is ironic that the AMNH had no part in the excavation or preparation of what is arguably its most
861 iconic fossil specimen, Barnum Brown having reunited it after a diaspora to three different
862 museums. It doubly ironic that, having acquired the skeleton, the museum did not even display it for
863 several years, then removed it from display after little more than a decade. Subsequent developments
864 show the value of palaeoart as a hypothesis-generating machine: there would be no rearing
865 *Barosaurus* mount without Greg Paul’s painting *Ambush at Como Creek*. It is striking that Dingus and
866 Gaffney independently had the idea for a rearing mount, despite their distaste for speculative
867 palaeobiology, each of them expecting the other to object — and that Norell, who they both
868 expected to veto the idea, was also on board. And it is satisfying that the biomechanical feasibility of
869 rearing diplodocines has subsequently been demonstrated mathematically. The mount certainly
870 achieved its stated goals of provoking not just public discussion, but much further scientific work.

871

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1094 **Figure Captions**

1095 **Figure 1.** The mounted skeleton of *Barosaurus lentus* AMNH 6341 in the Theodore Roosevelt
1096 Rotunda of the American Museum of Natural History, New York, as it appeared after the initial
1097 mounting and before repairs to the middle of the neck (see text). Photograph provided by
1098 Department of Vertebrate Paleontology Archive, Copyright © AMNH.

1099 **Figure 2.** Earl Douglass, aged 60, at the Carnegie Quarry in August 1922. He is seen here standing
1100 in front of the left femur of *Diplodocus* specimen #355 (now USNM 10865), with the sacrum,
1101 partial left ilium, last five dorsal vertebral centra and some ribs also in view (compare Figure 3, in
1102 which the femur is labelled 23, the ilium 24 and the last five dorsals 7–11). The *Diplodocus* is seen
1103 in ventral view with anterior to the top. Photograph uploaded to Wikimedia Commons by Ken
1104 Carpenter.

1105 **Figure 3.** Carnegie Quarry map showing the relative positions of two iconic sauropod specimens:
1106 #355 (*Diplodocus*, right) and #340 (*Barosaurus*, left, highlighted in pink, initially thought to be a
1107 second *Diplodocus*). Modified from Gilmore (1932:figure 1). The *Diplodocus* specimen would
1108 become USNM 10865, and the *Barosaurus* would become AMNH 6341. Note that in this map
1109 the cervical vertebrae of *Barosaurus* #340 are drawn to look like those of *Diplodocus*, which they
1110 were originally assumed to represent. Note also that the number of cervical vertebrae illustrated
1111 is ten, compared with the preserved count of nine, weakly corroborating the possibility that there
1112 was originally an additional, more anterior vertebra (C7 of present usage).

1113 **Figure 4.** *Barosaurus* AMNH 6341, cervical vertebrae C8–C16 in dorsal (where available) and
1114 lateral views, to scale. Lateral views from McIntosh (2005: fig. 2.1) except except C13, which we
1115 were able to photograph in collections. Extensive image manipulation was necessary to bring out
1116 the information in the dorsal view photographs, due to to poor photography conditions. C16 is
1117 sheered to the right, so the aspect is slightly left dorsolateral rather than true dorsal. C8 is on

display in the gallery with these vertebrae, but the structure of the display makes it impossible to photograph in dorsal view. C13 is on a shelf in collections, apart from the other cervicals, and we were not able to photograph it in dorsal view.

Figure 5. The Jurassic Hall in April 1939, photographs taken during or shortly after the renovations. **A.** The mounted skeleton of “*Brontosaurus*” (now thought to be *Apatosaurus*); in cabinets behind it the presacral vertebrae of *Barosaurus* AMNH 6341 can be seen in right lateral view. Note that the *Brontosaurus* mount has only thirteen cervicals, perhaps following Marsh’s (1891: plate XVI) skeletal reconstruction. **B.** The cabinets from the background of part A, showing the presacral sequence in anterodorsal view. In front of the vertebrae lie the right humerus, its posterior face uppermost and its proximal end facing the camera (left); and the right scapulocoracoid, its lateral face uppermost and its humeral glenoid roughly articulating with the humerus (right). Cropped from photographs 315932 (part A) and 315930 (part B) from the AMNH Research Library Digital Special Collections, by Charles H. Cole.

Figure 6. Public gallery exhibit panel for AMNH 6341, prepared around 1939 to accompany the vertebral sequence, scapulocoracoid and humerus depicted in Figure 5. Photographed by Mick Ellison, AMNH. Original caption reads:

Jurassic Period, Morrison Formation. Dinosaur National Monument, Vernal, Utah.

Barosaurus is related to Diplodocus and is characterized by extremely long cervical vertebrae which are more hollow than in other sauropods. Bones of skeleton, colored brown in restoration, are displayed below and in adjoining case; missing parts outlined. Restoration of skeleton, one-twelfth natural size. Dimensions of skeleton as restored — length 78 ft., height 19–1/2 ft. Acquired by purchase and exchange. Amer. Mus. No. 6341.

The representation of which bones are included in the specimen is a good match for our modern reconstruction (Figure 15), but there are some differences in the reconstruction. Most notably, only 14 cervical vertebrae are depicted, with the nine preserved cervicals interpreted as C6–C14. This is surprising as the closely related *Diplodocus* had been known with some confidence since Hatcher's (1901) monograph to have 15 cervicals. Also surprisingly, the 1939 reconstruction shows only nine dorsal vertebrae, compared with the ten described by Holland (1906:251–252) for *Diplodocus*. While McIntosh's catalogue of material indicates that the first 29 caudals are preserved, the 1939 diagram shows 30 caudals, of which the neural spines of Ca1–3 and Ca26–28 are absent and Ca21 is represented only by the anterior portion of its centrum. Finally, the 1939 diagram suggests that of the right dorsal ribs, only those of D7 and D8 are present, which contradicts McIntosh's (2005:57) assessment that two of the six preserved ribs are the left and right ribs of D1 or D2.

Figure 7. *Ambush at Como Creek*, painted by Gregory S. Paul in the late 1970s or early 1980s. In this revised version, completed in 1985, a pack of *Allosaurus* menace a herd of *Diplodocus*. While most of them, including a juvenile and two subadults, try to escape, one adult faces the attacking allosaurs in a threatening rearing posture. This painting was part of the inspiration for the AMNH's rearing *Barosaurus* mount. Copyright © Gregory S. Paul, 2022. Reproduced by kind permission.

Figure 8. The signage for the exhibit as it currently appears in the Roosevelt Memorial Hall. The headings read “Picturing the Past”, “*Barosaurus* Defends Her Young”, “Did This Really Happen?”, “How Were These Dinosaur Skeletons Assembled?” and “Collecting the Evidence”. See text for full transcription of the “Did This Really Happen?” section.

Figure 9. Prototype model of the proposed display. This model is three feet (91 cm) tall from the bottom of the base to the top of the rearing *Barosaurus*, and the base is 44 inches (112 cm) long (Mick Ellison, pers. comm, 2022). Despite some notable differences — for example, the model

has a much more strongly S-curved neck than the full-size mount, and its forefeet lack digits — the model seems to have been followed fairly closely in composing the final mount, and remains a beautiful object in its own right. Photograph by Mick Ellison (AMNH).

Figure 10. The mold creation process at RCI. Lazlo Eger paints separator on the mold wall surrounding a posterior cervical vertebra of *Barosaurus lentus* AMNH 6341, here seen upside-down in right ventrolateral view with anterior to the left. The separator on the mold wall allows the rubber latex for the next mold section to be applied without adhering to the first mold section, so that the mold sections will separate easily. Photograph provided by Department of Vertebrate Paleontology Archive.

Figure 11. Two views of the trial mount executed in the parking lot behind the RCI workshop in Spring 1991. **A.** Right posterolateral view, with an RCI employee adjusting the anterior part of the neck. Note that two vertebrae — one very posterior, the other in the middle of the neck — are absent at this point. **B.** left anterolateral view, with three RCI employees working on the anterior part of the neck: one in a cherry-picker, the other two on a scissor lift. Note that the two missing cervical vertebra have been added, and that the posterior one is a different colour from the others. The ribs are a noticeably lighter colour than the rest of the skeleton, presumably because they had not been painted at this point. Photographs provided by Department of Vertebrate Paleontology Archive.

Figure 12. Part way through the trial mounting process in the parking lot behind the RCI workshop in Spring 1991. In the lift platform, Peter May sets the humerus in place while Jack McIntosh advises on its articulation and Rick Eyre (back to the camera) looks on. At ground level, Gene Gaffney, Mark Norell and Lowell Dingus (left to right) observe. Photographs provided by Department of Vertebrate Paleontology Archive.

1189 **Figure 13.** Elements of the cast *Barosaurus* skeleton on the floor of the Roosevelt Memorial Hall of
1190 the American Museum of Natural History before mounting. Centrally, the skull and cervical
1191 vertebrae 3–16, all articulated in a single piece, with the atlas and axis still to be fitted behind the
1192 skull. By the wall to the right are the two halves of the 17th cervical vertebrae: the left half,
1193 highlighted in red, with anterior towards the camera; and the right half, highlighted in blue, with
1194 anterior facing away. Between these and the neck is the distal (whiplash) portion of a tail: possibly
1195 that of the main *Barosaurus*, but more likely that of the juvenile based on the small size of the
1196 most proximal caudals. Between this and the neck is a section that is probably the proximal part
1197 of the *Allosaurus* tail. Finally, intruding on the image from the left is the distal (whiplash) portion
1198 of the other *Barosaurus* tail, probably that of the adult. Photograph provided by Department of
1199 Vertebrate Paleontology Archive, Copyright © AMNH.

1200 **Figure 14.** Two views of the final mounting in the Roosevelt Memorial Hall of the American
1201 Museum of Natural History, both showing the use of two frighteningly unstable-looking cherry-
1202 picker cranes. The RCI employees in the higher crane are moving the head-and-neck assembly
1203 into place, while the two in the lower crane attempt to fit it to the anterior torso. Photographs
1204 provided by Department of Vertebrate Paleontology Archive, Copyright © AMNH.

1205 **Figure 15.** Skeletal reconstruction of *Barosaurus lentus* based primarily on AMNH 6341. Modified
1206 to show bones preserved in AMNH 6341 in white, and bones absent from this specimen (which
1207 had to be cast or modelled from other specimens for the mount) in grey. Some guesswork was
1208 involved here: for example, McIntosh (2005:43) says that six ribs are present, but it is not possible
1209 to say which six ribs. Base image copyright © 2022 Scott Harman, all rights reserved. Used by
1210 kind permission.

1211 **Figure 16.** The skull used in the mounted *Barosaurus*. **A.** Cranium to rear, mandible to the front,
1212 both in left dorsolateral view, photographed in 1991. **B.** Cranium to rear, mandible to the front,
1213 both in right dorsolateral view, photographed in 1991. This skull was copied from that of the

mounted Carnegie *Diplodocus*. Note its similarity to the skull “as placed in the restoration at the British Museum” in Holland (1906:figure 1). **C.** The skull used in the Paris mount of *Diplodocus*, held by an unnamed worker, in left lateral view, photographed in 1908. That of the *Barosaurus* mount is identical.

Figure 17. The anterior part of the AMNH *Barosaurus* neck (skull and first eight cervicals, though the atlas and axis are absent at this point) during the mounting process. **A.** lying on the floor, seen in roughly dorsal view. **B.** held upright, seen in right lateral view. Note that the vertebrae in positions 3–7 are those of *Diplodocus*, with relatively high bifurcated neural spines, while the vertebra in position 8 is that of *Barosaurus*, with a low and unbifurcated spine. Comparison with the Carnegie *Diplodocus* shows that those in positions 3–6 are C3–C6 of *Diplodocus*, but that the 8th cervical vertebra of *Diplodocus* is used in position 7 to give a more even progression of increasing length. Photographs provided by Department of Vertebrate Paleontology Archive, Copyright © AMNH.

Figure 18. *Barosaurus Defends Her Young*, John Gurche’s painting of a rearing *Barosaurus* confronting an *Allosaurus* and protecting a juvenile, created especially for the publicity materials for the mount when unveiled in 1991. This painting was used in gallery signage and in numerous publications (e.g. the cover of *New York Newsday* for 29 November 1991; the cover of the AMNH’s own magazine *Natural History* for December 1991).

Figure 19. Lego model of the rearing AMNH *Barosaurus*, created by Ambrosino (a pseudonym). Reproduced with permission. This model is built from the pieces of Lego’s official Dinosaur Fossils kit (number 21320) and its instructions are freely available (Ambrosino 2023). These instructions have been used as the basis for a commercially available, but unauthorized, kit.

1237 **Figures**

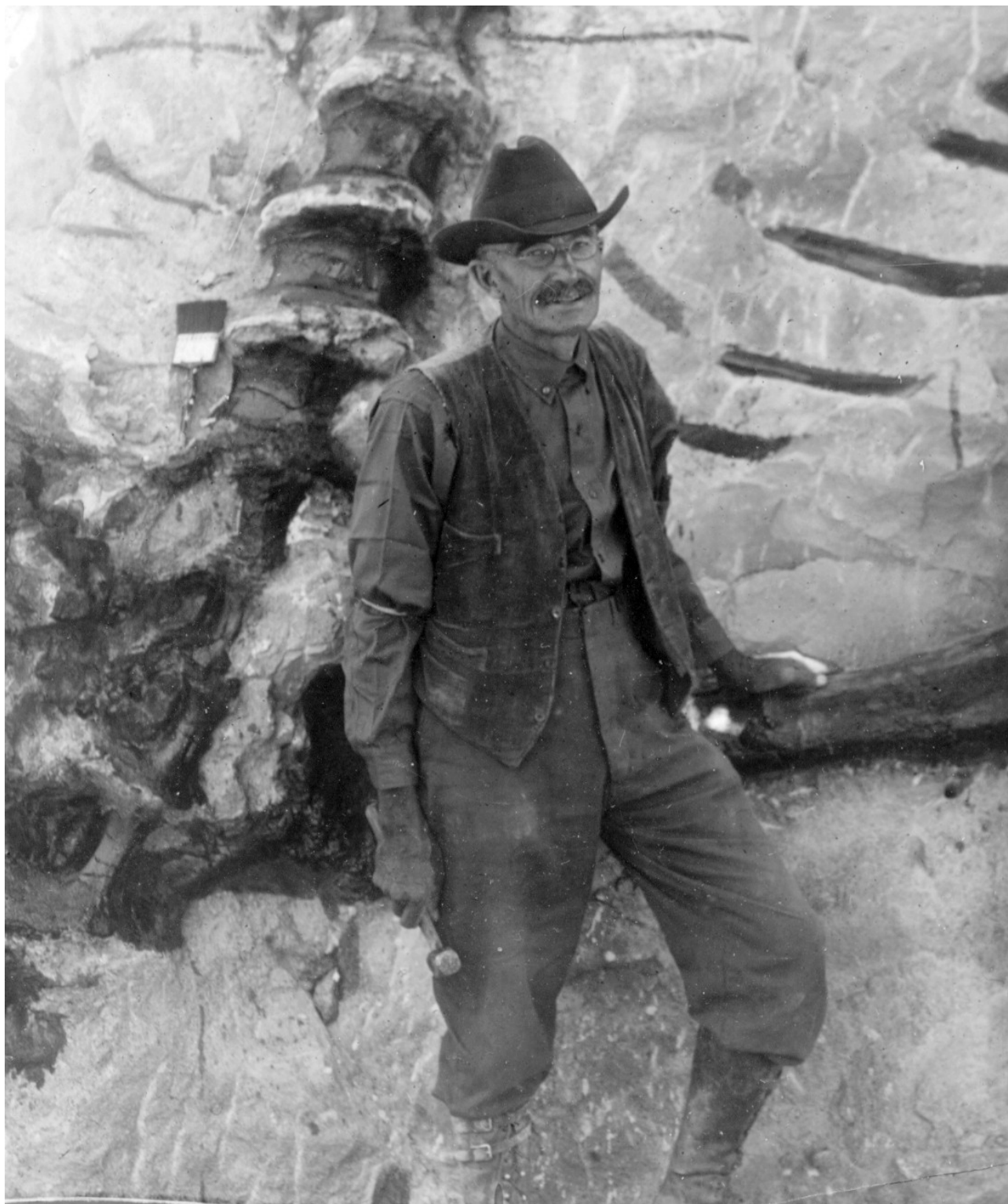
1238 **Figure I**



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1241 **Figure 2**



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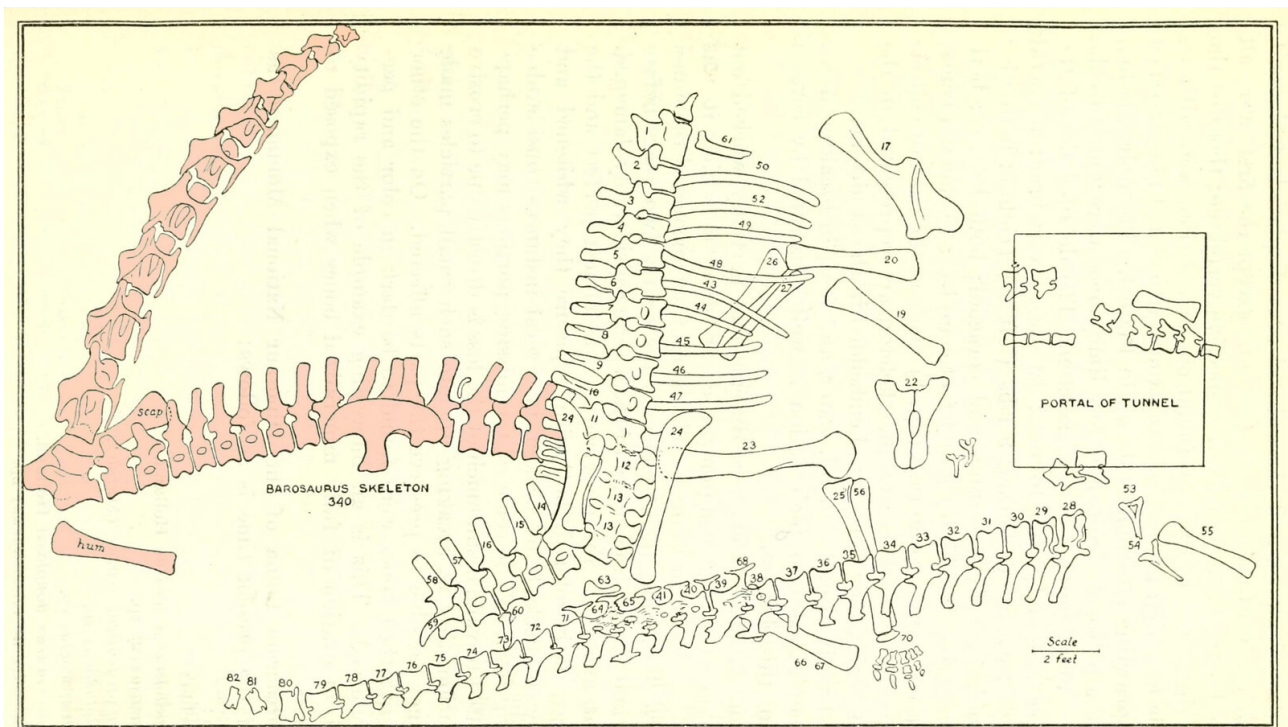
1244 **Figure 3**

FIGURE 1.—Diagram or quarry map showing the relative positions of the bones of the *Diplodocus* skeleton as it was embedded in the sandstone. Nos. 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10, dorsals 1 to 10, respectively; 11, 12, and 13, sacral vertebrae; 14, 15, and 16, caudal vertebrae 1, 2, and 3; 17, right scapula and coracoid; 19, left humerus; 20, right humerus; 22, pubes; 23, left femur; 24, ilia; 25, left tibia; 26, right ulna; 27, right radius; 28, sixth caudal vertebra; 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, seventh to nineteenth caudal vertebrae; 43, 44, 45, 46, 47, sixth to tenth ribs of the left side; 48, fifth rib of left side; 49, fourth rib of left side; 50, third rib of left side; 52, second rib of left side; 53, fourth chevron; 55, left ulna; 56, left fibula; 57, fourth caudal vertebra; 58, fifth caudal vertebra; 59, chevron; 60, third chevron; 63, metatarsal; 64, twenty-first caudal vertebra; 65, twentieth caudal vertebra; 66, 67, ischia; 68, chevron; 70, left hind foot; 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, caudal vertebrae; 82, caudal vertebra not used in mounted skeleton

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1247 **Figure 4**



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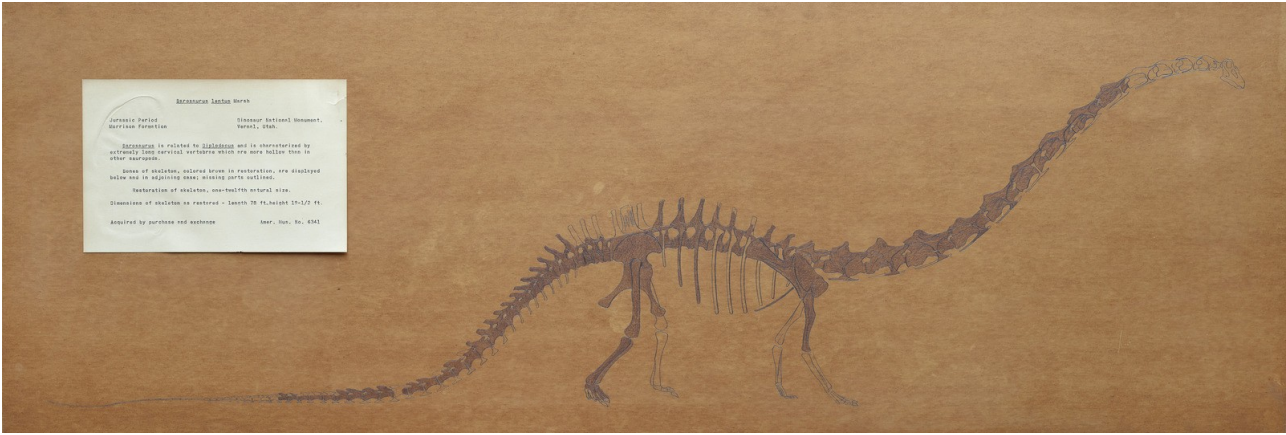
1250 **Figure 5**



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1253 **Figure 6**



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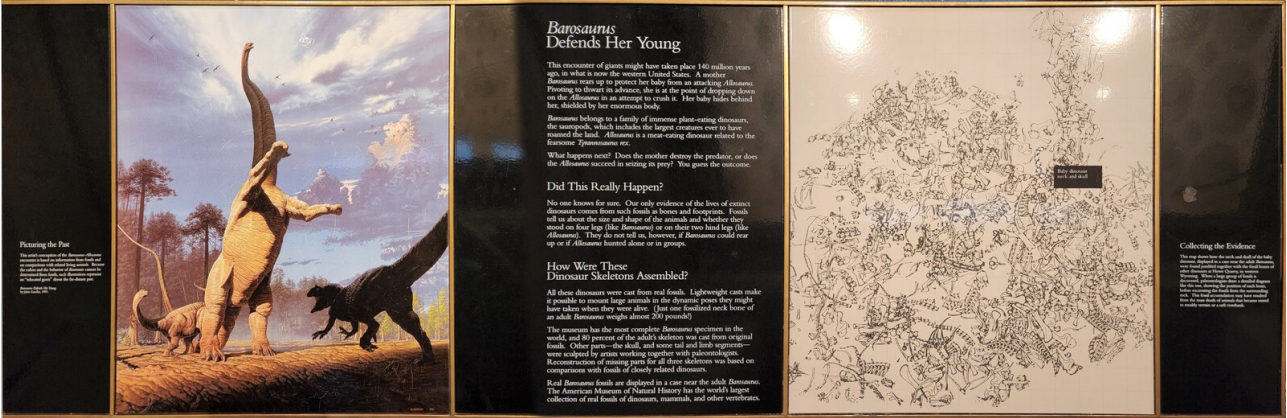
1256 **Figure 7**



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1259 **Figure 8**



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1262 **Figure 9**



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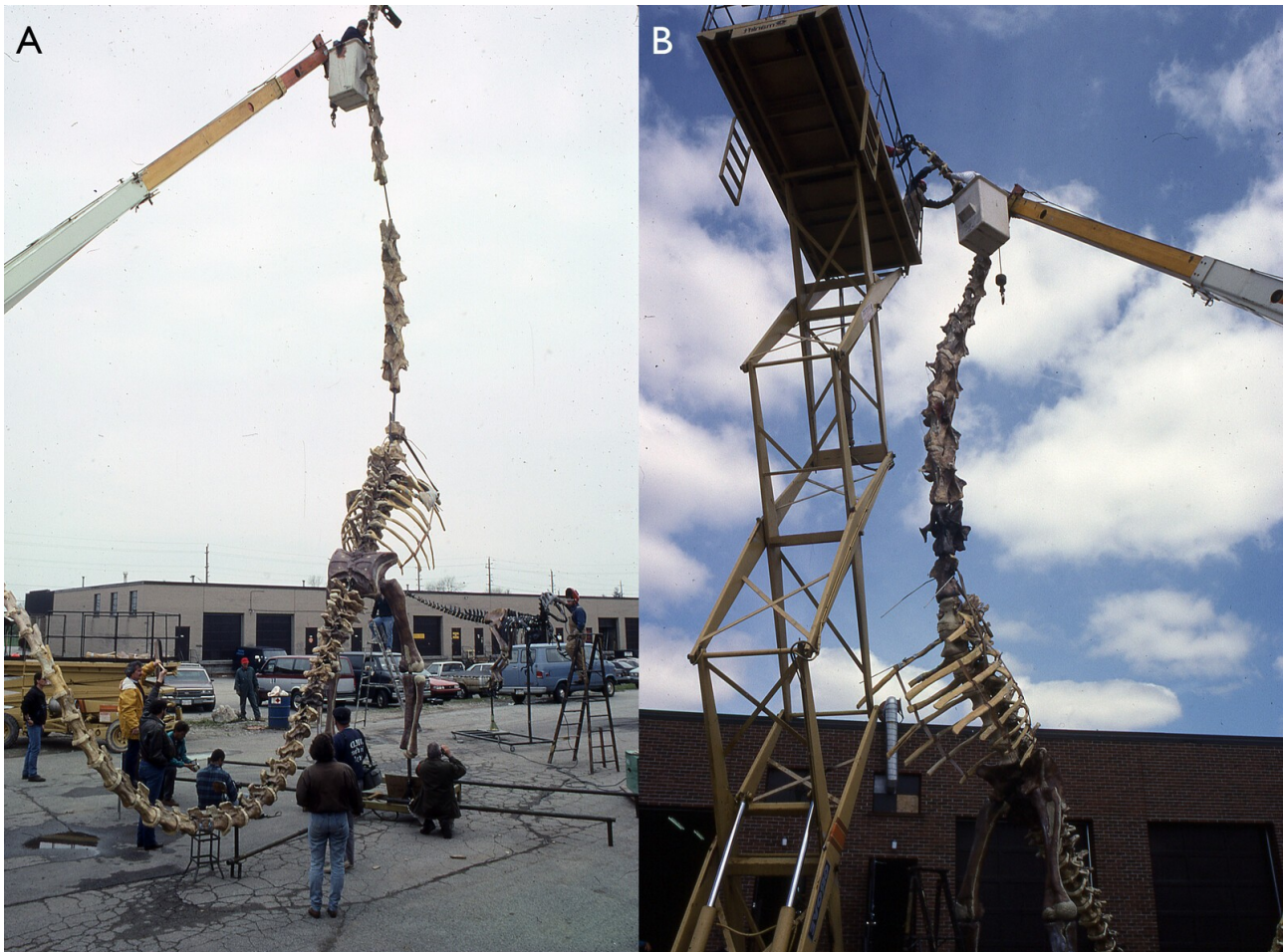
1265 **Figure 10**



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1268 **Figure 11**



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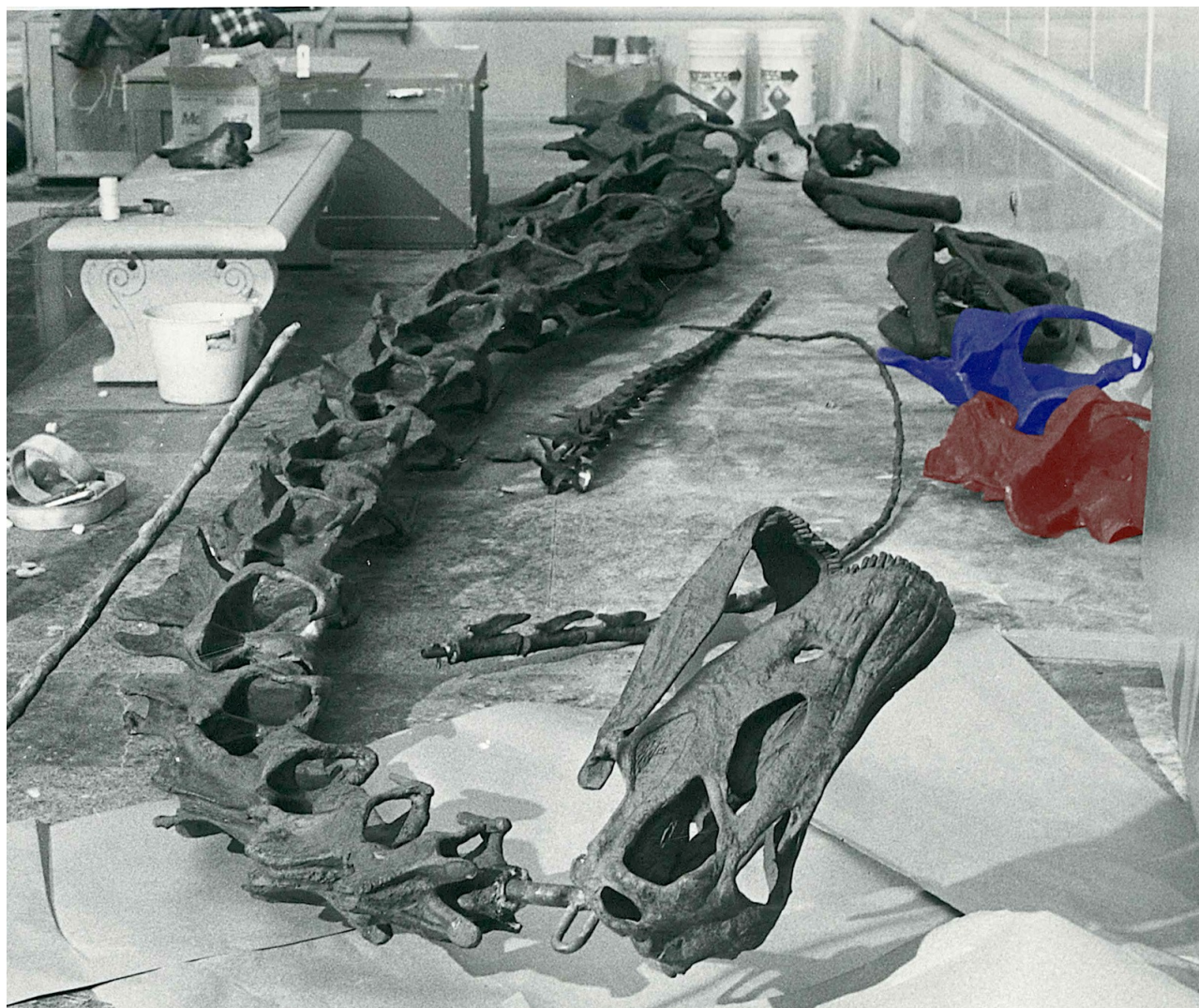
1271 **Figure 12**



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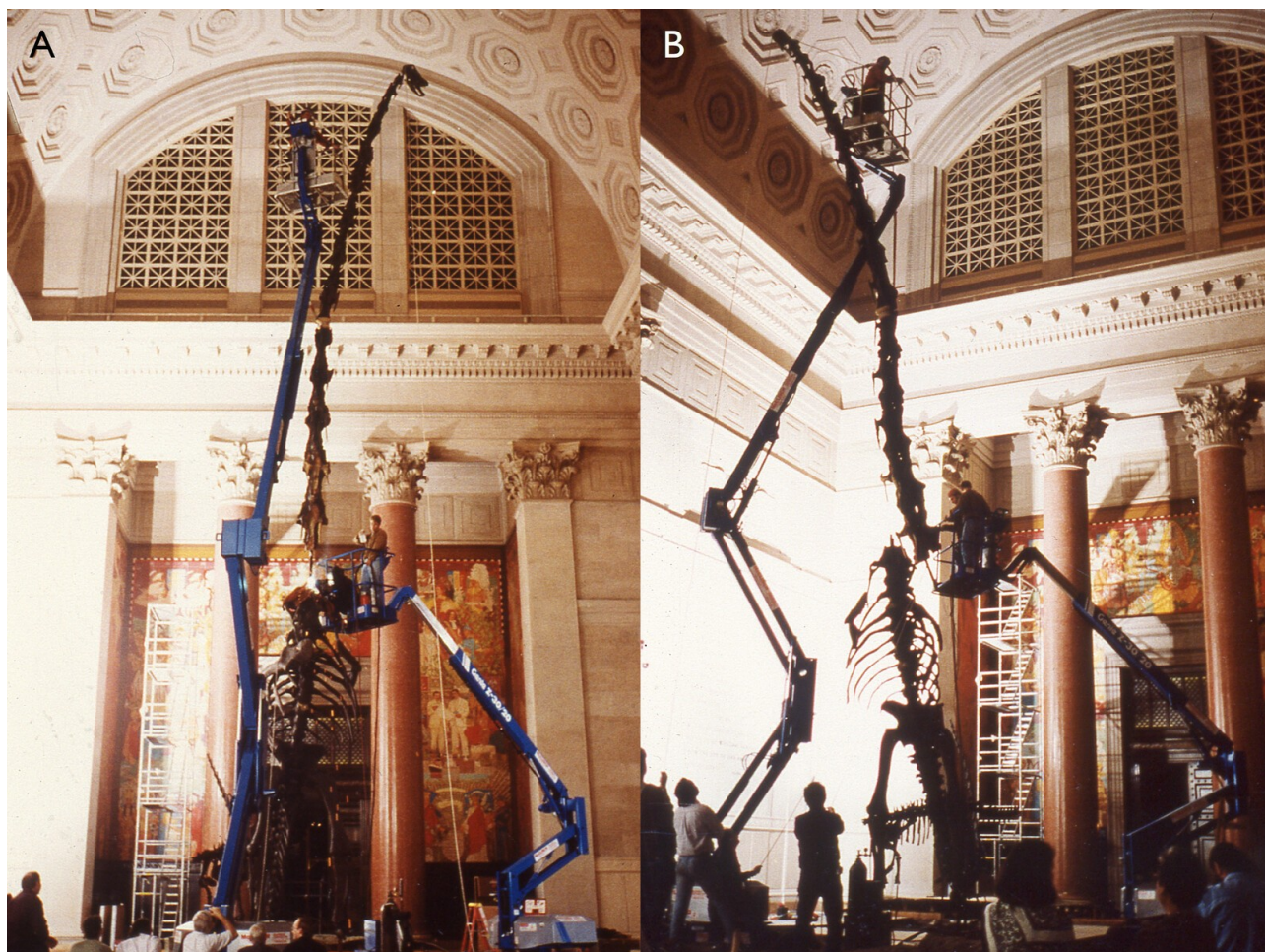
1274 **Figure 13**



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1277 **Figure 14**



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1280 **Figure 15**

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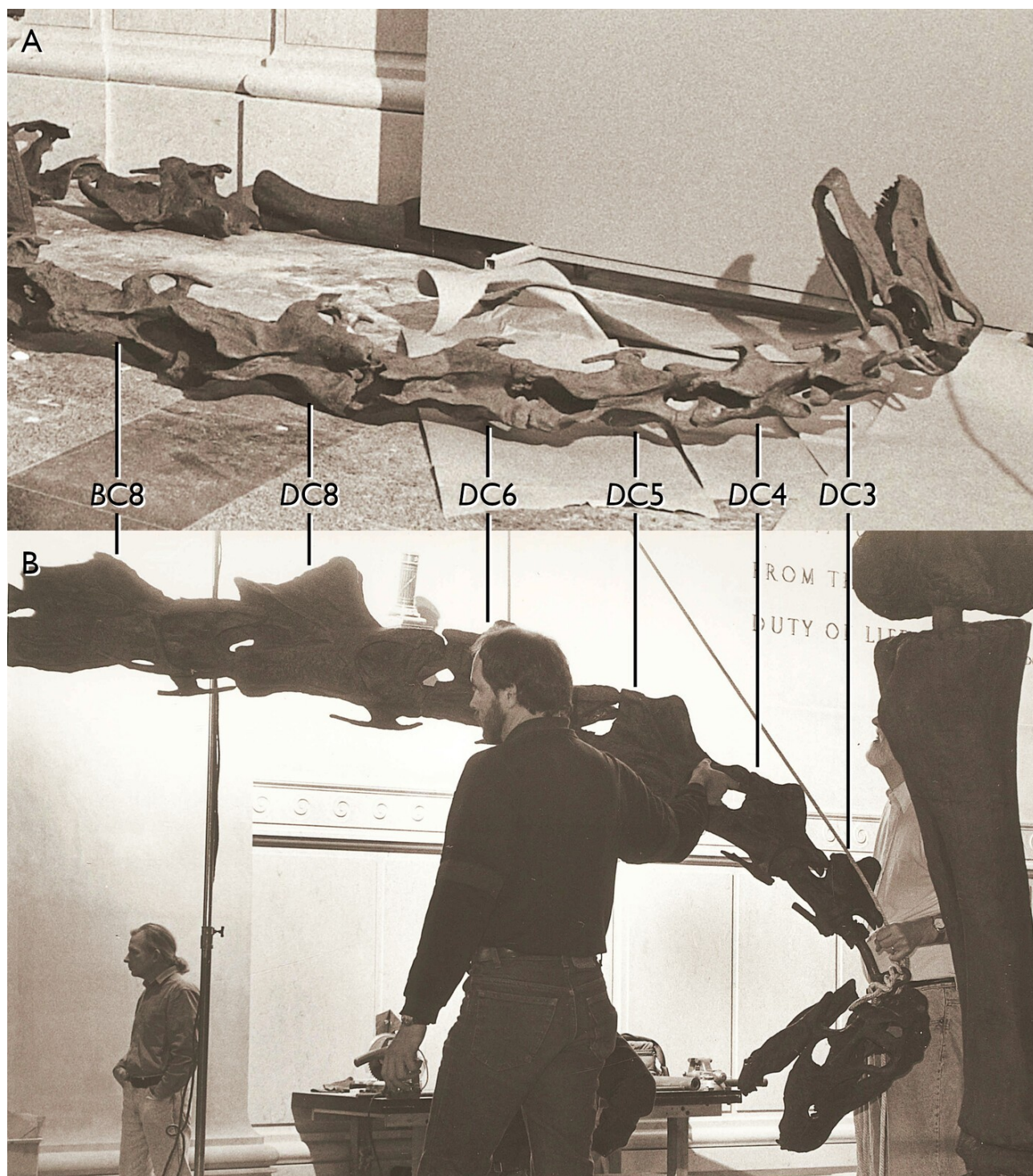


1283 **Figure 16**



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1286 **Figure 17**

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1292 **Figure 19**



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1295 **Tables**1296 **Table I**1297 Summarised history of the genus *Barosaurus* and the rearing skeleton mount AMNH 6341.

Date	Event
1880s	The first fossils of what is now <i>Barosaurus</i> were discovered in the 1880s by Mrs. E. R. Ellerman on land owned by Mrs. Rachel Hatch half a mile east of Piedmont on the eastern rim of the Black Hills of South Dakota.
Summer 1889	O. C. Marsh visited the site with J. B. Hatcher, and collected part of the tail for the Yale Peabody Museum (YPM).
1890	Marsh (1890) very briefly described and named the new genus and species <i>Barosaurus lentus</i> based on six caudal vertebrae and a chevron from the initial excavation.
Late August 1898	Marsh sent George Wieland to collect the rest of the skeleton.
1899	First published scientific support for the plausibility of rearing: Osborn (1899:213) wrote that the tail of <i>Diplodocus</i> “functioned as a lever to [...] raise the entire forward portion of the body upwards.”
1907	Charles R. Knight painted a rearing <i>Diplodocus</i> .
Late 1916	The <i>Barosaurus lentus</i> holotype YPM 429 was fully prepared; Richard S. Lull gave a presentation of the specimen at the eighth annual meeting of the Paleontological Society in Albany, NY.
1917	Lull’s published abstract very briefly described the type specimen.
1919	Lull’s detailed monographic description of the type specimen.

Date	Event
1922	Janensch (1922:464) erroneously referred diplodocid material from the Tendaguru Formation of Tanzania to <i>Barosaurus</i> as the new species <i>B. africanus</i> (now <i>Tornieria africana</i>).
1922–1923	Two diplodocine skeletons were partially excavated from the Carnegie quarry (now Dinosaur National Monument). One of these, designated #340 was thought at that time to be <i>Diplodocus</i> . Earl Douglass took most of it to the University of Utah, but parts of its tail were sent to the Carnegie Museum.
1923	The neck of #340 was sent to the USNM to provide material for a <i>Diplodocus</i> skeletal mount based on the other diplodocine #355 (USNM 10865). When prepared, the neck sent to the USNM proved to belong not to <i>Diplodocus</i> but to <i>Barosaurus</i> .
1929	Barnum Brown realised that the USNM neck and the Carnegie tail belonged to the same individual as the partial skeleton at the University of Utah. He negotiated to acquire all three portions of this skeleton for the American Museum of Natural History (AMNH) in New York.
1931	Work began on the Roosevelt Memorial Hall that is the main entrance to the AMNH. It was completed in 1936.
9 February 1933	The portion of the skeleton held by USNM was belatedly shipped to AMNH. The reunited skeleton was given the specimen number AMNH 6341.
17 April 1939	The presacral vertebrae, scapulocoracoid and humerus of AMNH 6341 were exhibited in the Hall of Early Dinosaurs.
Early 1950s	The AMNH 6341 elements were removed from display as part of Edwin H.

Date	Event
	Colbert's renovation of the galleries.
1952	The original Carnegie <i>Diplodocus</i> molds, having been used ten times by the Carnegie museum to create plaster casts from museums around the world, were donated to the Utah Field House of Natural History in Vernal, Utah.
8 June, 1957	A concrete <i>Diplodocus</i> cast made from these molds was unveiled outside the Utah Field House.
1970	Eugene S. Gaffney succeeded Colbert as curator of the fossil reptile collection at the AMNH.
1977	Peter May began working with skeletal mounts at the Royal Ontario Museum.
1978	Gregory S. Paul painted <i>Ambush at Como Creek</i> , a widely reproduced piece that depicts rearing <i>Diplodocus</i> individuals under attack from <i>Allosaurus</i> .
1986	The AMNH began planning an extensive renovation of its fossil halls.
1989	Jim Madsen's company Dinolab unmounted the concrete <i>Diplodocus</i> of Vernal, used it to create second-generation molds, and created a new, lightweight cast for the Utah Field House.
1990	Peter May went full time with his company Research Casting International (RCI).
28 March 1990	A formal agreement was made between AMNH and RCI for the project to mount a cast of <i>Barosaurus</i> AMNH 6341 along with a juvenile <i>Barosaurus</i> and an <i>Allosaurus</i> .
Fall 1990	RCI took the repaired and cleaned <i>Barosaurus</i> fossils from New York to

Date	Event
	Toronto.
Late 1990/early 1991	RCI created casts of the <i>Barosaurus</i> elements and filled in the gaps with second generation Carnegie <i>Diplodocus</i> casts from Dinolab.
Spring 1991	A test erection of the rearing <i>Barosaurus</i> mount was carried out in Toronto.
Summer 1991	Gene Gaffney and Peter May found a site in northeastern Montana that could be molded and cast to provide a base for the skeletons.
November 1991	The work in Toronto was complete, and the completed exhibit was transported to New York.
29 November 1991	The John Gurche painting of a rearing <i>Barosaurus</i> , prepared for the exhibit, appeared on the front cover of <i>New York Newsday</i> .
3 December 1991	The completed exhibit — rearing adult and hiding juvenile <i>Barosaurus</i> , and running <i>Allosaurus</i> — was unveiled to select guests.
4 December 1991	The exhibit was opened to the public.